





Reliability

Embedded Systems

Lirida Alves de Barros-Naviner
Master Program



Outline

Dependability
- Introduction
- Deterministic model
- Probabilistic models

Fault and defect tolerance improvement

Fault tolerance assessment

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Dependability

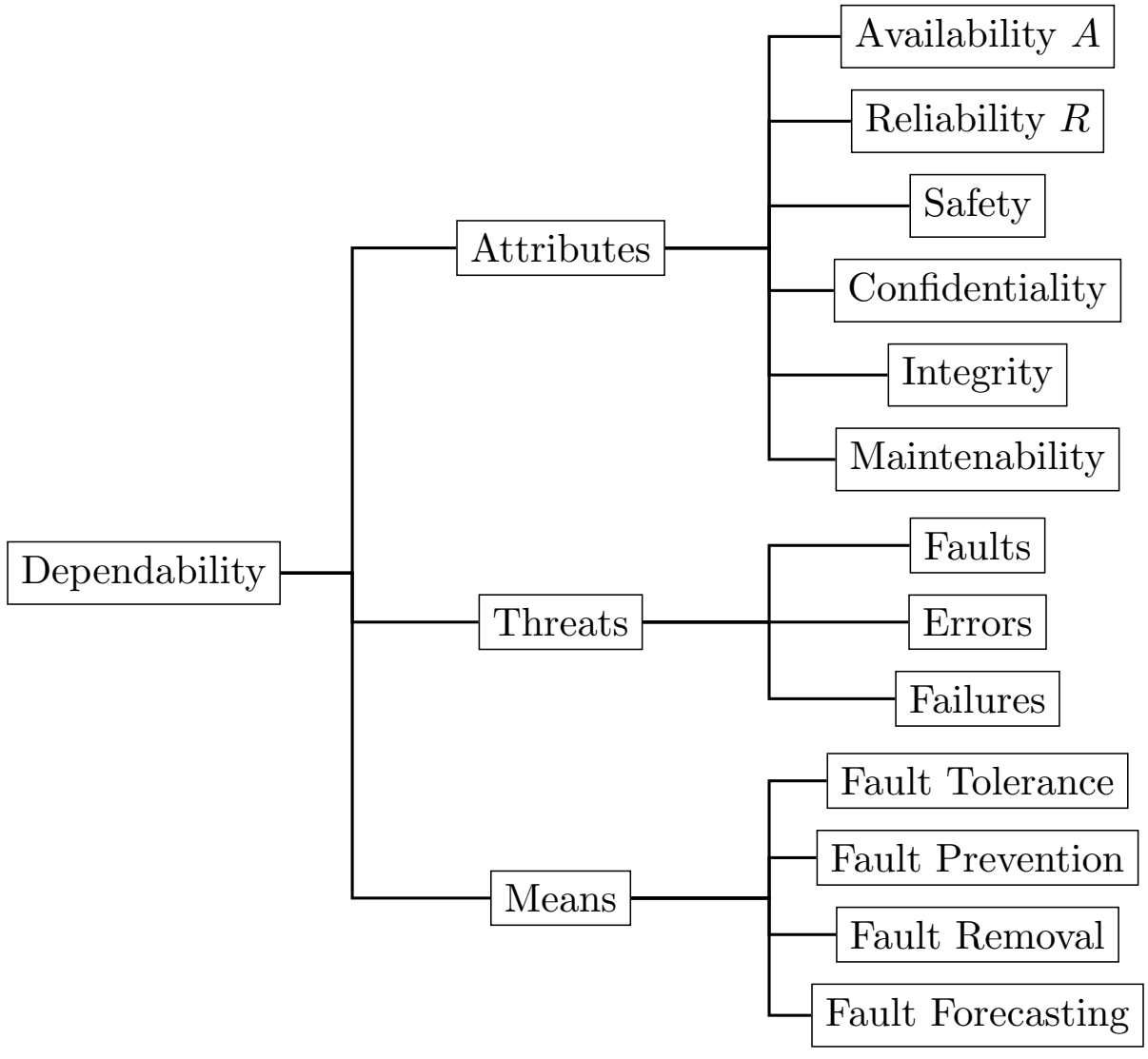
Definition

Dependability is the ability of a system to deliver service that can *justifiably* be trusted.

Definition

Dependability is the ability of a system to avoid *service failures* that are *more frequent or more severe* than is *acceptable*.

Taxonomy



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Dependability Attributes

- **Availability:** readiness for correct service.
- **Reliability**: continuity of correct service.
- **Safety**: absence of catastrophic consequences on the user(s) and the environment.
- **Integrity**: absence of improper system alterations.
- **Maintainability**: ability to undergo modifications and repairs.

Dependability Threats

- **Fault**: an *unexpected (incorrect) condition* that can lead the system to achieve *abnormal states*. A fault can lead to an error.
- **Error**: an *undesired (incorrect) state* of the system. An error can lead to an incorrect response of the system.
- **Failure**: an *incorrect response* of the system. It means the service provided by the system differs from the expected one.

Means to Ensure Dependability

- **Fault prevention**: avoid things go wrong!
- **Fault tolerance**: deal with, when things go wrong!
- **Fault removal**: make it right, if things went wrong!
- **Fault forecasting**: be aware of how wrong things can go

Failure Rate

Definition

The **failure rate** λ is the expected number of failures per unit time.

- For a system with n components λ can be estimated as:

n independent components

$$\lambda = \sum_{i=1}^{n} \lambda_i$$

Mean Time to Failure

Definition

The **Mean Time to Failure (MTTF)** of a system is the expected time of the occurrence of the first system failure.

n components

$$\text{MTTF} = \frac{1}{n}\sum_{i=1}^{n} t_i$$

Failures In Time

$$\text{FIT} = \frac{10^9}{MTTF}$$

Mean Time to Repair

Definition

The **Mean Time to Repair (MTTR)** of a system is the average time required to repair the system.

- MTTR is often given in terms of the repair rate μ, which is the expected number of repairs per unit of time

$$\text{MTTR} = \frac{1}{\mu}$$

Availability

Definition

Instantaneous availability $A(t)$ is the probability the system operates at time t.

- **Interval availability** stands for the average of $A(t)$ over a mission period:

$$\mathrm{A(T)} = \frac{1}{T}\int_0^T A(t)dt$$

- **Steady-state availability** applies when availability is time independent:

$$\mathrm{A}(\infty) = \lim_{T\to\infty} A(T) = \frac{n \times MTTF}{n \times MTTF + n \times MTTR} = \frac{\mu}{\mu + \lambda}$$

  - Supposes n failures during lifetime

Mean Time Between Failures

Definition

The **Mean Time Between Failures (MTBF)** is the average time between failures of the system.

$$\text{MTBF}=\text{MTTF}+\text{MTTR}$$

Assuming repair makes the item perfect

$$\text{MTBF}=\frac{MTTF}{A(\infty)}$$

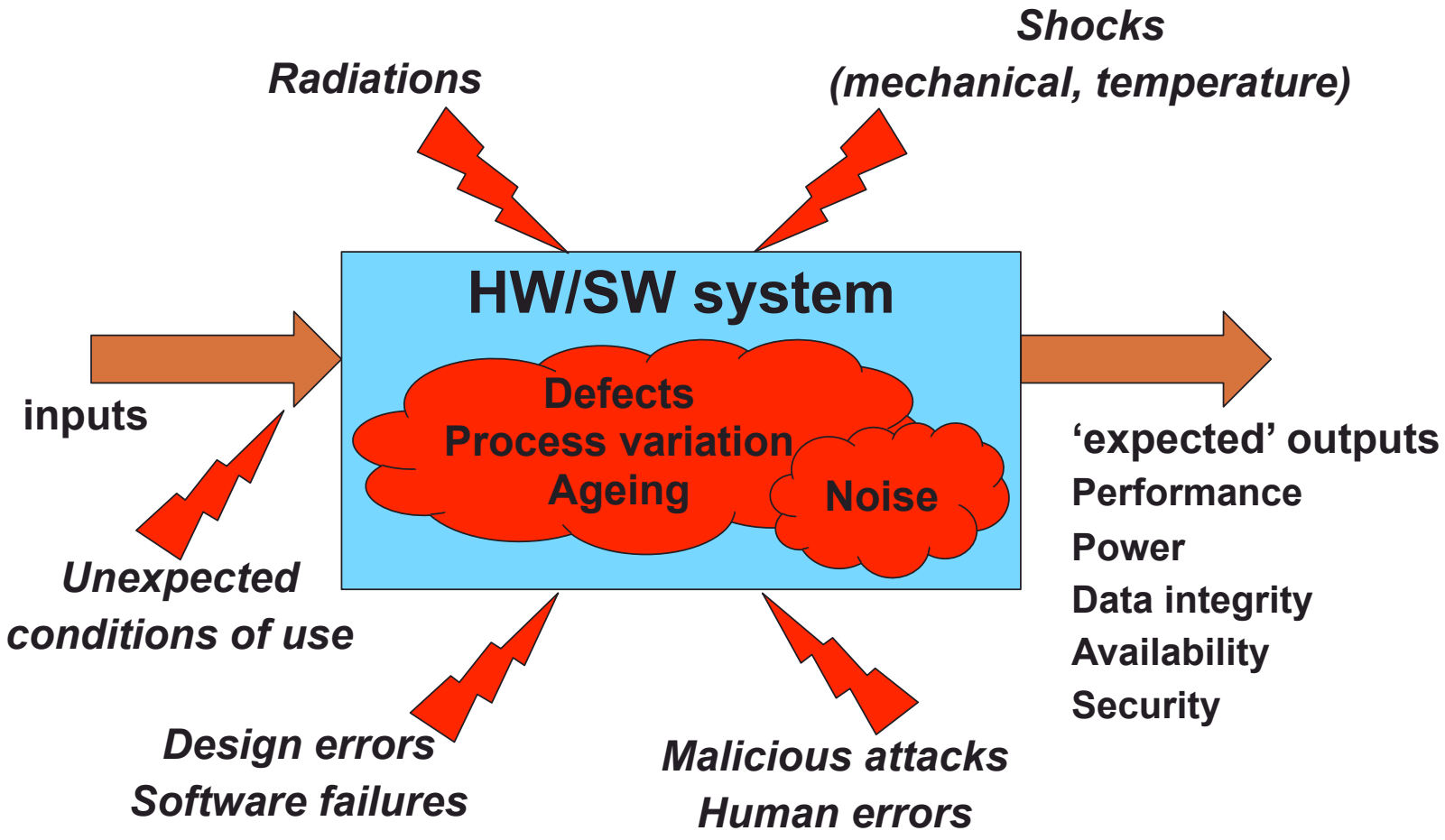
Fault Coverage

Definition

The **Fault Coverage** FC is the conditional probability related to expected actions when faults occurs.

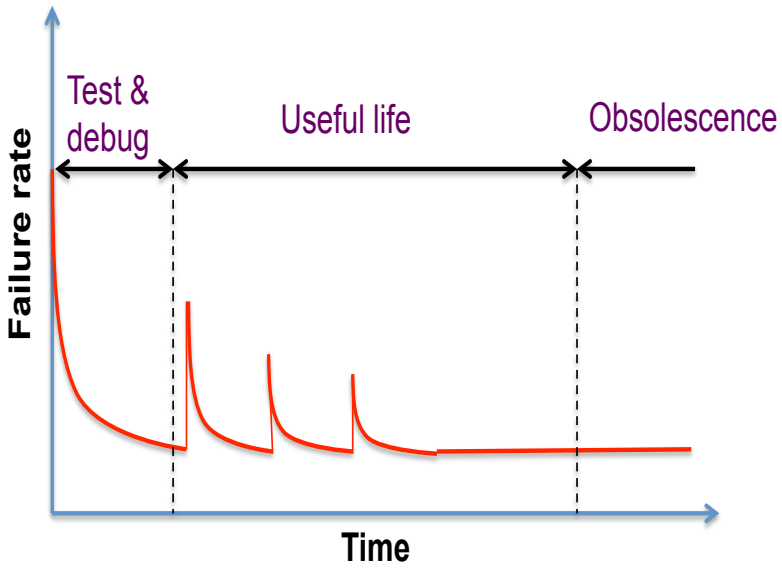
- FC= P(detected faults | existent faults)
- FC= P(located faults | existent faults)
- FC= P(recovered faults | existent faults)
- FC= P(contained faults | existent faults

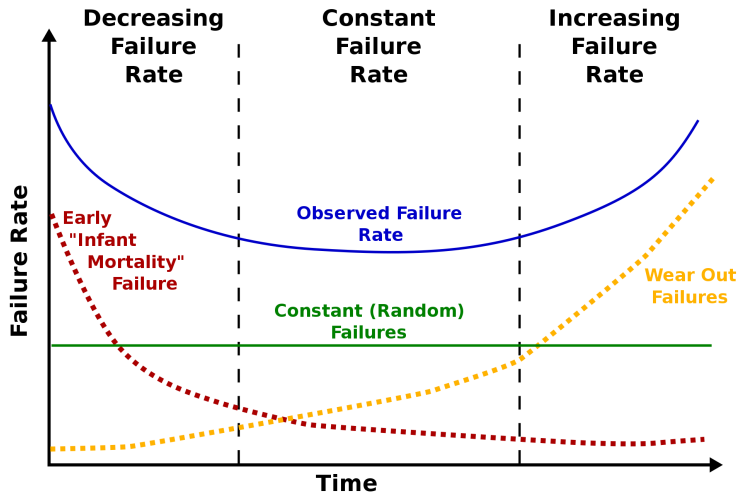
What About Embedded Systems?



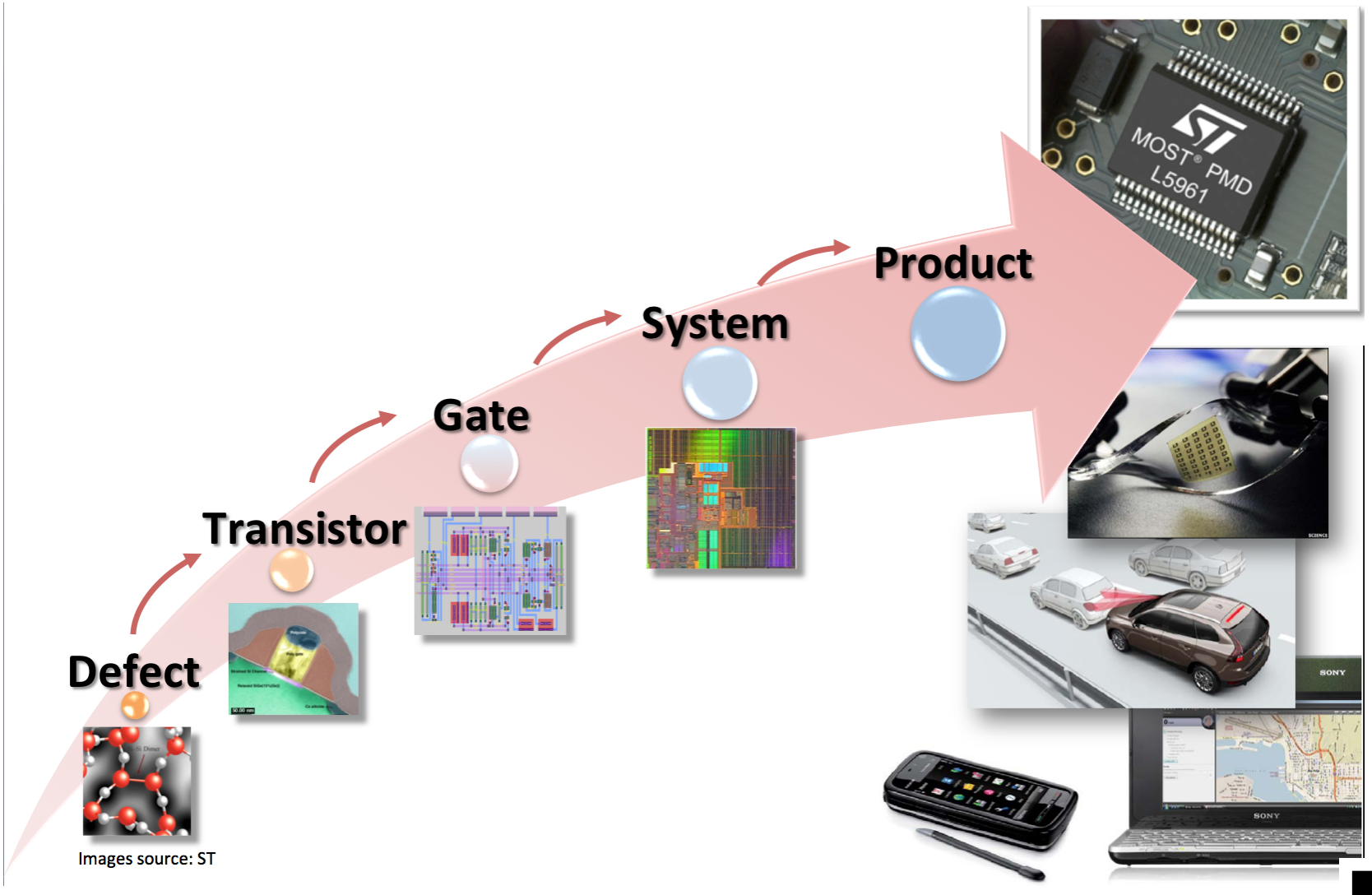
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SW and HW Faults



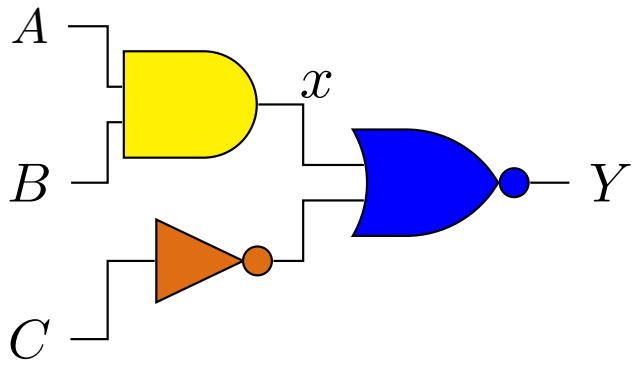


Default/Fault Propagation



Images source: ST

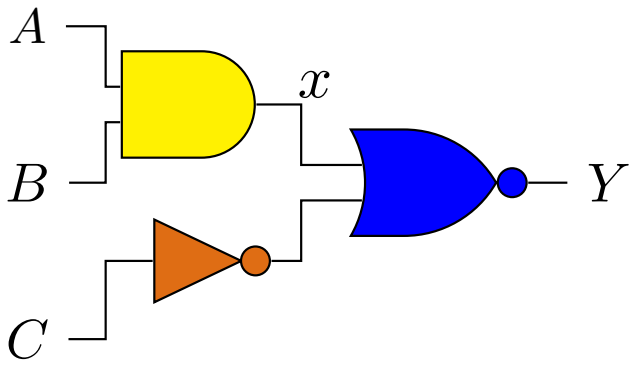
Fault Models: Bit-flip and Stuck-at



A	B	C	x	Y
0	0	0	0	0
0	0	1	0	1
0	1	0	0	0
0	1	1	0	1
1	0	0	0	0
1	0	1	0	1
1	1	0	1	0
1	1	1	1	0

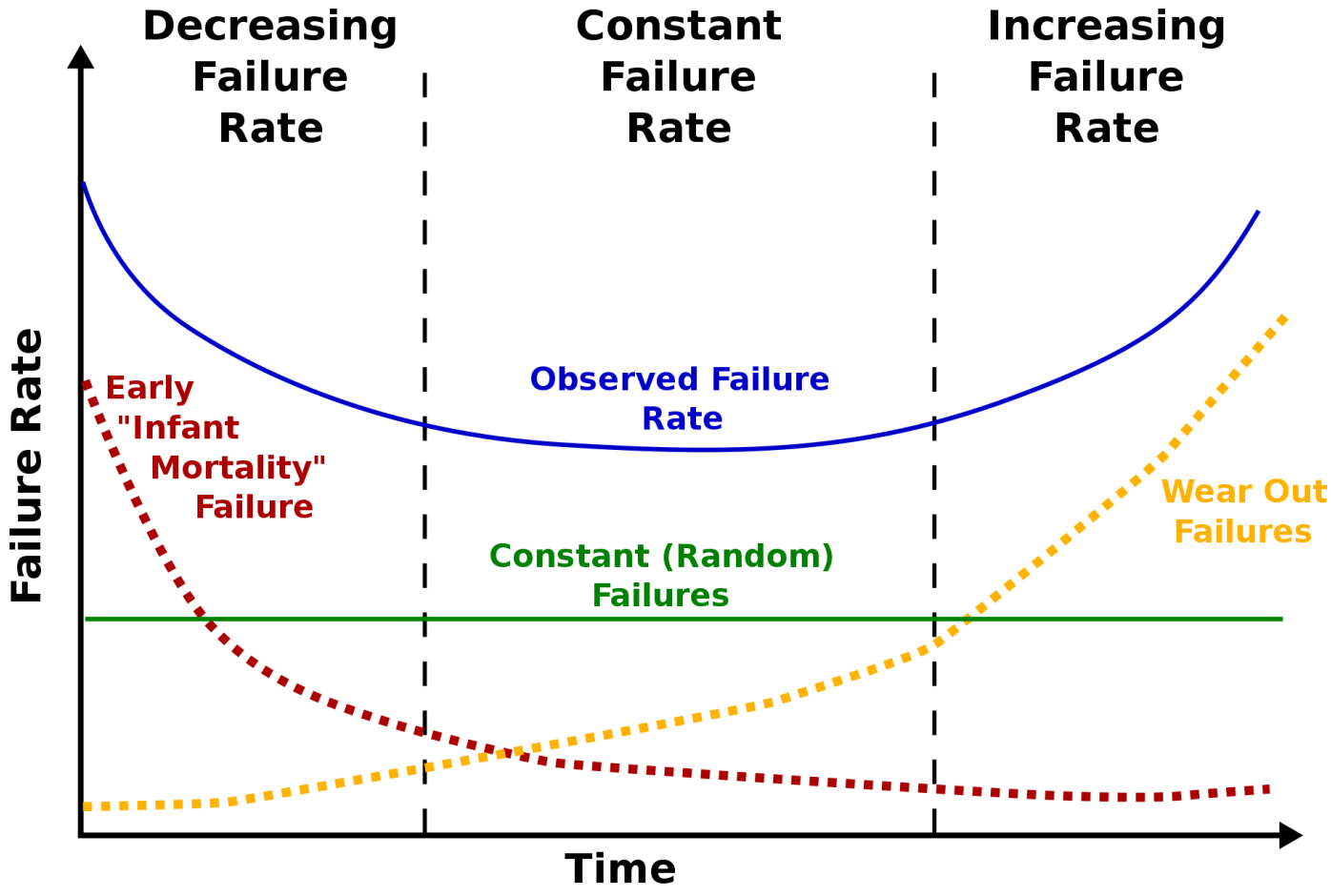
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Fault Models: Bit-flip and Stuck-at



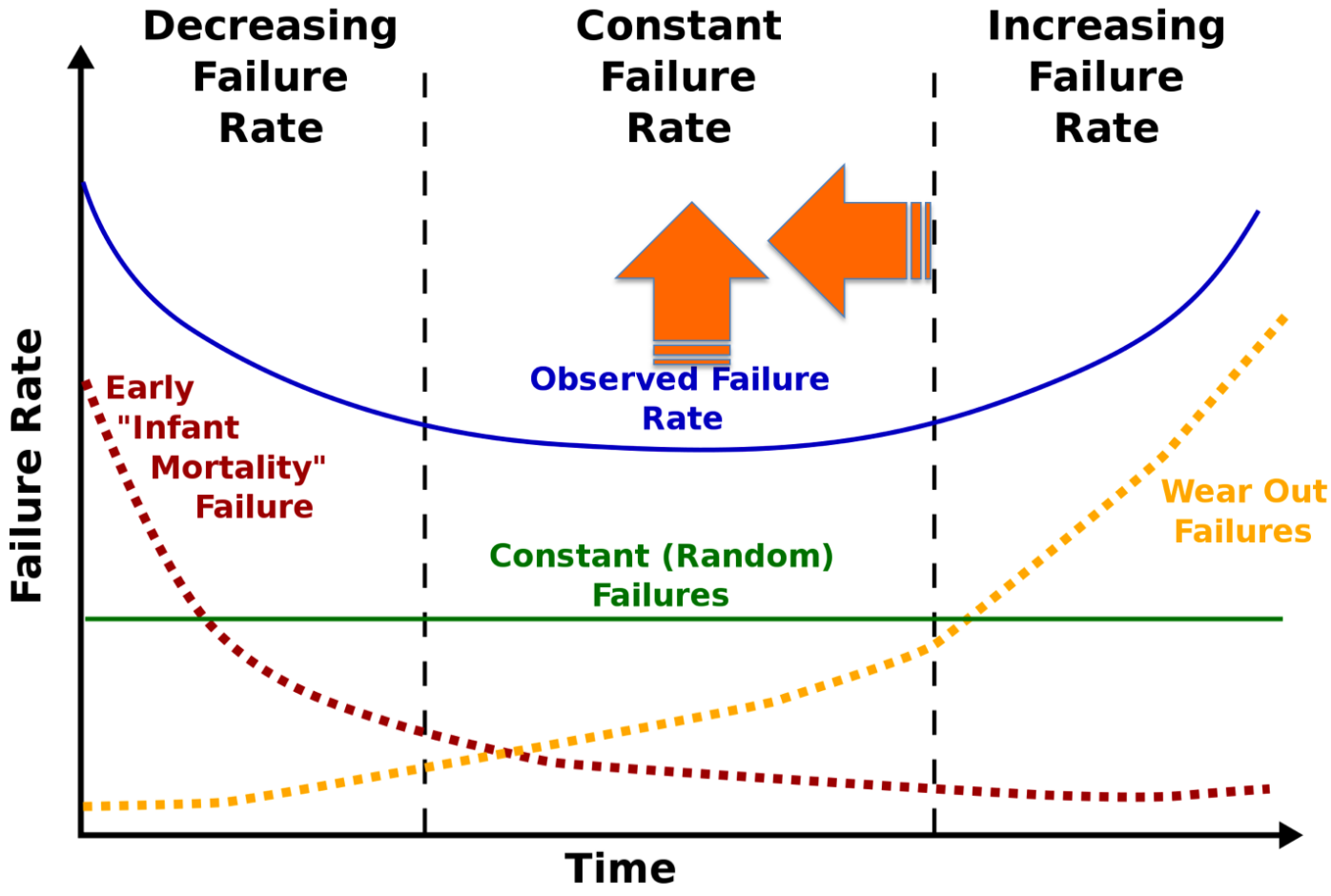
A	B	C	x	Y
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0	1	0	1	0
0	1	1	1	0
1	0	0	1	0
1	0	1	1	0
1	1	0	1	0
1	1	1	1	0

CMOS Scaling and Reliability



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CMOS Scaling and Reliability



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Fault tolerance assessment

Prior to Beginning

- We focus on system modeling
- We consider the system consists of several components: $c_1, c_2, \cdots, c_n$
- We look for a function that enables reliability analysis

Deterministic Model

Definition

The **state of a component** c_i is defined as

$$x_i = \begin{cases} 0 & \text{if the component } c_i \text{ is not fonctionning} \\ 1 & \text{if the component } c_i \text{ is functionning} \end{cases}$$

Definition

The **state set** is defined as the vector composed by the components states

$$\mathbf{x} = (x_1 x_2 \cdots x_n)$$

Deterministic Model (cont.)

Definition

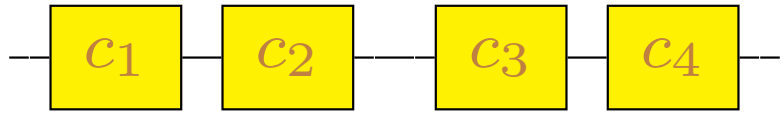
The **system state** is defined as

$$\xi(\mathbf{x}) = \begin{cases} 0 & \text{if the system is not fonctionning with state set } \mathbf{x} \\ 1 & \text{if the system is functionning with state set } \mathbf{x} \end{cases}$$

Reliability Block Diagram

- Static representation (no reference to time)
- Each component represented by a block
- Based on logic (Boolean algebra)
- Independence of components failures
- Behavior facing faults represented by the connections between blocks

Series System

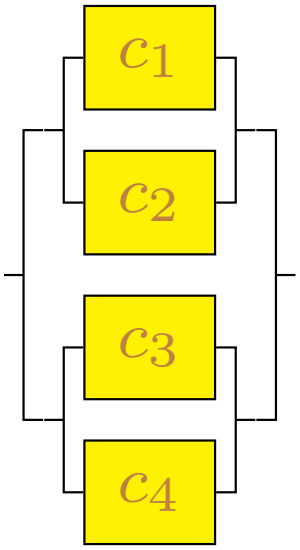


$$\xi(\mathbf{x}) = \begin{cases} 0 & \text{if there exists an } i \text{ such that } x_i = 0 \\ 1 & \text{if } x_i = 1 \text{ for all } i \in [1; n] \end{cases}$$

$$= \prod_{i=1}^{n} x_i$$

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Parallel System



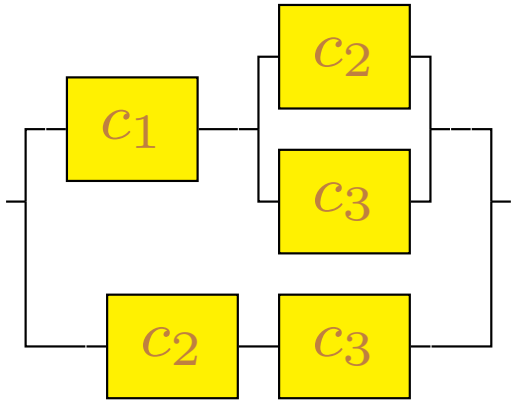
$$\xi(\mathbf{x}) = \begin{cases} 0 & \text{if } x_i = 0 \text{ for all } i \in [1;n] \\ 1 & \text{if there exists an } i \text{ such that } x_i = 1 \end{cases}$$

$$= 1 - \prod_{i=1}^{n}(1 - x_i)$$

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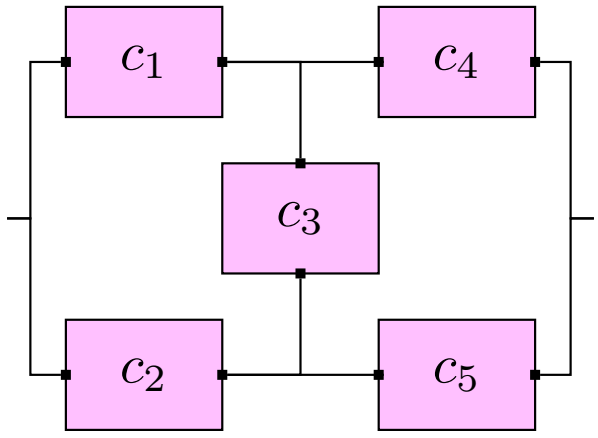
Combined Series-Parallel System

Example: 2 out of 3 structure



$$\xi(\mathbf{x}) = \begin{cases} 0 & \text{if } \sum_{i=1}^{n} x_i < k \\ 1 & \text{if } \sum_{i=1}^{n} x_i \geq k \end{cases}$$

Non Series-Parallel System



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Coherent System

Definition

A system of n components is **coherent** if its function $\xi(\mathbf{x})$ is nondecreasing in $\mathbf{x}$ and there are no irrelevant components.

Definition

A function $\xi(\mathbf{x})$ is **nondecreasing** in $\mathbf{x}$ if $\xi(x_1 \cdots x_{i-1}\mathbf{0}x_{i+1} \cdots x_n) \leq \xi(x_1 \cdots x_{i-1}\mathbf{1}x_{i+1} \cdots x_n)$.

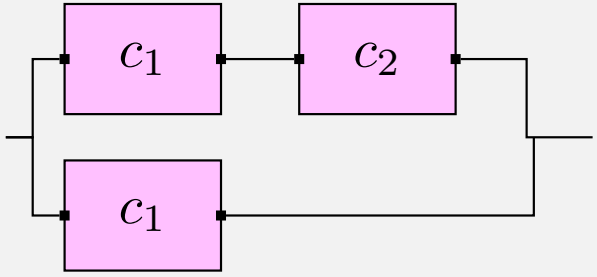
Definition

A component c_i is **irrelevant** if its state x_i has no impact on the function $\xi(\mathbf{x})$.

Coherent System (cont.)



A non coherent structure:



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Structural Importance

Definition

The **structural importance** of a component c_i in a coherent system of n components is

$$I_\xi(i) = \frac{1}{2^{n-1}} \sum [\xi(1_i, \mathbf{x}) - \xi(0_i, \mathbf{x})]$$

Path Vector

Definition

A **path vector** for a coherent system is a vector $\mathbf{x}$ such as $\xi(\mathbf{x}) = 1$.

Definition

A **minimal path** for a coherent system is a path vector $\mathbf{x}$ such as $\xi(\mathbf{y}) = 0$ for all $\mathbf{y} < \mathbf{x}$.

Definition

Given two vectors $\mathbf{x}$ and $\mathbf{y}$, $\mathbf{x} < \mathbf{y}$ if and only if $x_i \leq y_i$ for $i = 1, 2, \cdots, n$ and $x_i < y_i$ for some i.

Definition

A **minimal path set** P_j for a coherent system is a set with all components associated to a given minimal path vector.

Cut Vector

Definition

A **cut vector** for a coherent system is a vector $\mathbf{x}$ such as $\xi(\mathbf{x}) = 0$.

Definition

A **minimal cut vector** for a coherent system is a cut vector $\mathbf{x}$ such as $\xi(\mathbf{y}) = 1$ for all $\mathbf{y} > \mathbf{x}$.

Definition

A **minimal cut set** C_j for a coherent system is a set with all components associated to a given minimal cut vector.

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Probabilistic Model

Definition

The **random state of a component** c_i is defined as

$$X_i = \begin{cases} 0 & \text{if the component } i \text{ has failed} \\ 1 & \text{if the component } i \text{ is functionning} \end{cases}$$

Definition

The **random state of the set of components** in a system is defined as

$$\mathbf{X} = (X_1 X_2 \cdots X_n)$$

Component and System Reliability

Definition

The **reliability of a component** c_i is defined as the *probability* that component c_i is functionning [at prescribed time]

$$R_i = P\{X_i = 1\} = q_i$$

Definition

The **reliability of a coherent system** is defined by

$$R = P\{\xi(\mathbf{X}) = 1\}$$

Reliability Importance

Definition

The **reliability importance of a component** c_i in a coherent system of n components is given by

$$I_{R_i} = \frac{\partial R(\mathbf{q})}{\partial q_i} = R(1_i, \mathbf{q}) - R(0_i, \mathbf{q})$$

for $i = 1, 2, \cdots, n$

Reliability Bounds

Theorem

The reliability of a coherent system of n independent components respects

$$\prod_{i=1}^{n} q_i \leq R(\mathbf{q}) \leq 1 - \prod_{i=1}^{n} (1 - q_i)$$

Lifetime Models

Definition

Reliability is the ability of an item to perform its *required functions* under *stated conditions* and for a *specified period of time* (IEEE definition).

- A *item* or a *component* may mean a simple (i.e logic gate) or a complex system.
- The definition suggests *behaviour item evolution.*

Lifetime Representations

- We denote T a continuos nonnegative random variable that represents the **lifetime** of a item.
  - Note that *time* may stand to hours but also to number of flips, number of km, etc.
- We consider functions that define the distribution of T, representing the **failure time** of a item.

Probability Density Function

Definition

The **probability density function** (PDF) is defined as

$$f(t) = \lim_{\Delta t \to 0} \frac{P\{t \leq T \leq t + \Delta t\}}{\Delta t}$$

$$f(t) = 0 \text{ for } t < 0 \quad f(t) \geq 0 \text{ for } t \geq 0 \quad \int_0^1 f(t)dt = 1$$

- The PDF indicates the likelihood of failure for any t

Reliability (or Survivor) Function

Definition

The **reliability function** $R(t)$ is defined as

$$R(t) = R(\mathbf{q}, t) = \quad = \quad P\{T \geq t\} \quad \forall t \geq 0$$

$R(t)$ must be nonincreasing and respect $R(0) = 1$, $\lim_{t \to \infty} R(t) = 0$

Hazard Function

Definition

The **hazard function** $h(t)$ is defined as the amount of risk associated to an item at time t.

$$\text{h(t)}= \frac{f(t)}{R(t)}$$

$$\begin{aligned} h(t) &= \lim_{\Delta t \to 0} P\{t \leq T \leq t + \Delta t | T \geq t\} \\ &= \lim_{\Delta t \to 0} \frac{P\{t \leq T \leq t + \Delta t\}}{P\{T \geq t\}} \\ &= \lim_{\Delta t \to 0} \frac{R(t) - R(t + \Delta t)}{R(t)\Delta t} \\ &= \frac{f(t)}{R(t)} \end{aligned}$$

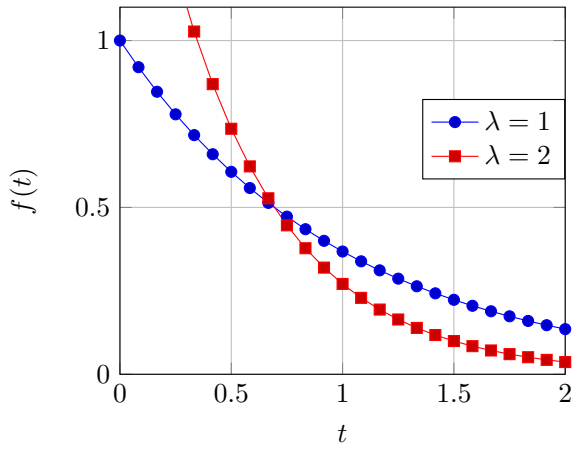
- $h(t)$ represents the instantaneous **failure rate**.
- $h(t)$ must respect $\int_0^\infty h(t)dt = \infty$, $h(t) \geq 0 \quad \forall t \geq 0$

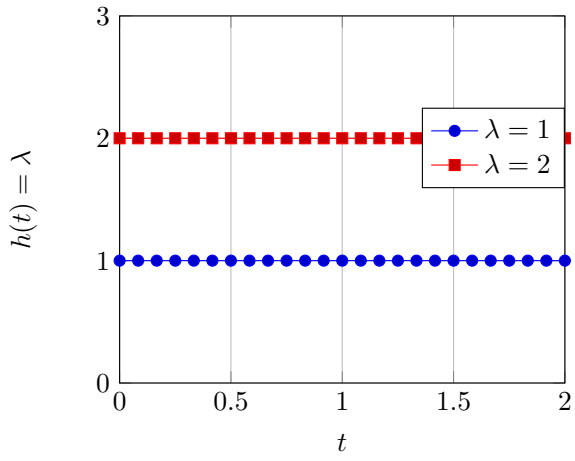
Lifetime Distributions

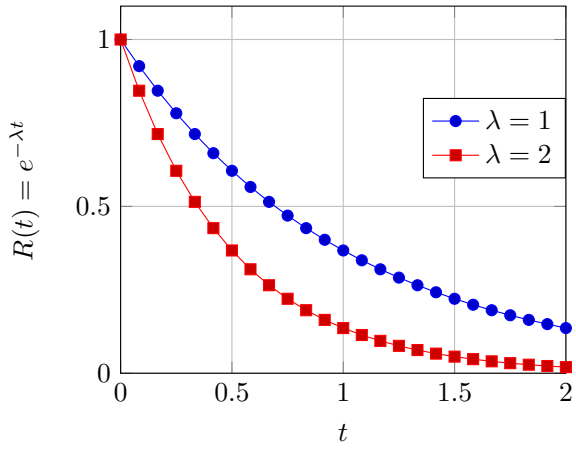
	Exponential	Weibull	Gamma
$R(t)$	$e^{-\lambda t}$	$e^{-(\lambda t)^{\kappa}}$	$1 - I(\kappa, \lambda t)$
$f(t)$	$\lambda e^{-\lambda t}$	$\kappa \lambda^{\kappa} t^{\kappa-1} e^{-(\lambda t)^{\kappa}}$	$\frac{\lambda}{\Gamma(\kappa)} (\lambda t)^{\kappa-1} e^{-\lambda t}$
$h(t)$	λ	$\kappa \lambda^{\kappa} t^{\kappa-1}$	$\frac{f(t)}{R(t)}$

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Exponential Distribution







Applies for useful life zone in bathtub curve

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Markov Chain

State	Time
Discrete	Discrete
Discrete	Continuous
Continuous	Discrete
Continuous	Continuous

Continuous Time Markov Chains (CTMC)

- Memoryless system
- Discrete space
- Exponential distribution (events at constant rates)

Markov Chain

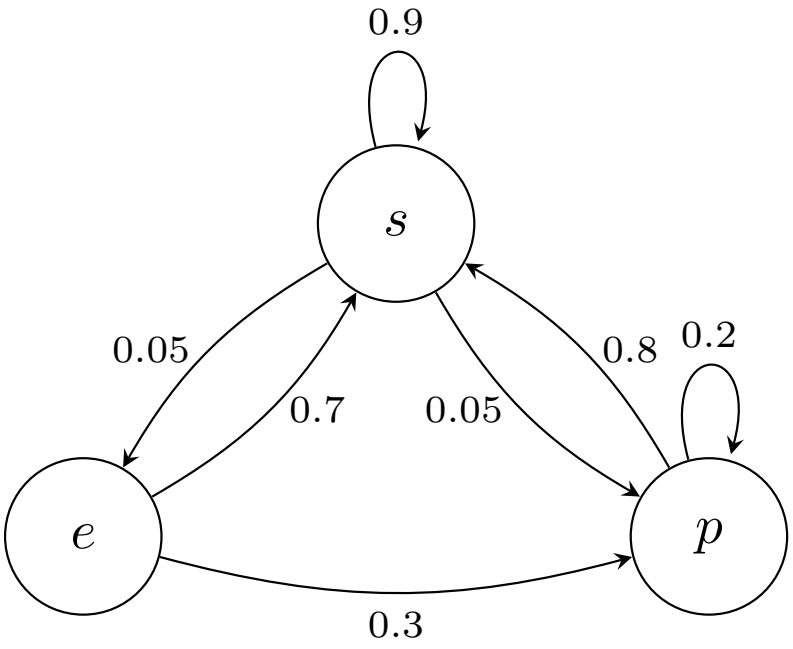


A lazy, gourmand, and lovely hamster

- When Doudou sleeps, there are 9 chances out of 10 that it will be lying in bed the next minute. When it wakes up, it climbs to its happiness, so there is 1 chance out of 2 that it will be playing and 1 chance out 2 it will be eating.
- Its meals last for one minute and then it starts to play (3 chances out of 10) or it goes to sleep (7 chances out of 10).
- Doudou gets tired quickly. Frequently it goes back to sleep (8 chances out of 10) but, as it loves its spinning wheel, sometimes it continues to play.

- Knowing that Doudou is sleeping now, what will it likely be doing in three minutes?

Markov Chain & Simulation Matrix



$$\mathrm{S}=\begin{bmatrix} 0.9 & 0.05 & 0.05 \\ 0.7 & 0 & 0.3 \\ 0.8 & 0 & 0.2 \end{bmatrix}$$

- There are three states: sleep (s), eat (e) and play (p)
- Each element $s_{i,j} \in S$ gives the probability of next state being j given that actual state is i

Simulation Matrix & Behavior

- $P(t) = \begin{bmatrix} P_s(t) & P_e(t) & P_p(t) \end{bmatrix}$ gives the probability of each state for a given time t
- Hypothesis: initial state is s, then
  - $P(0) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$
- Probability of next states are:
  - $P(1) = P(0).S = \begin{bmatrix} 0.9 & 0.05 & 0.05 \end{bmatrix}$
  - $P(2) = P(1).S = \begin{bmatrix} 0.885 & 0.045 & 0.07 \end{bmatrix}$
  - $P(3) = P(2).S = \begin{bmatrix} 0.884 & 0.04425 & 0.07175 \end{bmatrix}$
- Probability at time n: $\boxed{\mathrm{P(n)=P(n\text{-}1).S=P(0)S}^n}$

Markov Chain & Transition Matrix

$$P_i(t+dt) = P_i(t)\left[1 - \sum_{j \neq i} s_{i,j}(t)dt\right] + \sum_{j \neq i} P_j(t)s_{j,i}dt$$

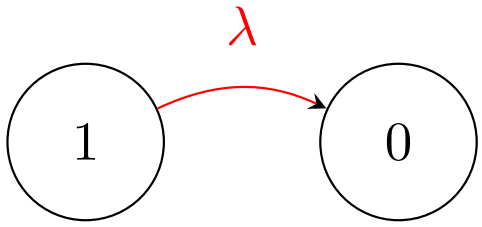
$$\frac{P_i(t+dt) - P_i(t)}{dt} = -P_i(t)\sum_{j \neq i} s_{i,j}(t)dt + \sum_{j \neq i} P_j(t)s_{j,i}dt$$

$$\frac{dP(t)}{dt} = M(t)P(t)$$

- M is the transition matrix. Each $m_{i,j} \in M$ gives the rate with sytem passes from state i to state j
  - $m_{i,j,i \neq j} = s_{j,i}$ and $m_{i,i} = \sum_{j \neq i} s_{j,i}$

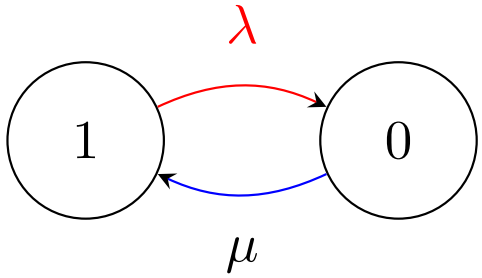
Markov Chain & Transition Matrix (cont.

- One component without repair



$$M = \begin{bmatrix} m_{11} & m_{10} \\ m_{01} & m_{00} \end{bmatrix} = \begin{bmatrix} -\lambda & \lambda \\ 0 & 0 \end{bmatrix}$$

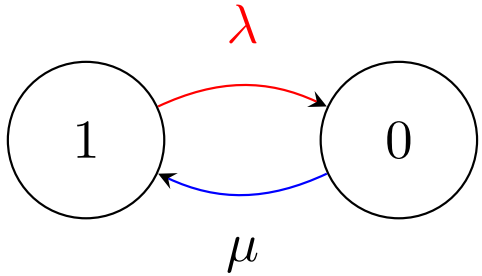
- One component with repair



$$M = \begin{bmatrix} m_{11} & m_{10} \\ m_{01} & m_{00} \end{bmatrix} = \begin{bmatrix} -\lambda & \lambda \\ \mu & -\mu \end{bmatrix}$$

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State Transition Equations (STE)



$$M = \begin{bmatrix} m_{11} & m_{10} \\ m_{01} & m_{00} \end{bmatrix} = \begin{bmatrix} -\lambda & \lambda \\ \mu & -\mu \end{bmatrix}$$

$$\begin{aligned} -\lambda P_1 + \mu P_0 &= 0 \\ \lambda P_1 - \mu P_0 &= 0 \\ P_1 + P_0 &= 1 \end{aligned}$$

$$\boxed{P_1 = \frac{\mu}{\lambda + \mu} \text{ and } P_0 = \frac{\lambda}{\lambda + \mu}}$$

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Reliability and STE

$$\mathrm{R(t)}=\sum_{i\in\mathcal{T}} P_i(t)$$

$$\mathrm{R(t)}=1-\sum_{i\in\mathcal{F}} P_i(t)$$

Assuming repair makes the item perfect, $\mathcal{T}$ is the set of fonctionning states, $\mathcal{F}$ is the set of failing states

Conclusions

- This course focuses on reliability, which is a dependability's attribute
  - Dependability is an essential quality metric for many systems
- The previous slides dealt with different methods for dependability analysis
- The reliability of digital electronics components has specific characteristics
  - Fault models, quality metrics, etc.
- The next slides will explore techniques for reliability improvement and reliability assessment

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Fault tolerance assessment

Digital Design for Reliability

How to design reliable processors on unreliable devices?



- Defect tolerance
- Fault tolerance
- Prevention
- Masking
- Detection
- Correction

Defect tolerance

Based on hardening the devices

- More strict design rules at mask-level
- More expensive manufacturing (area)

Based on programmability

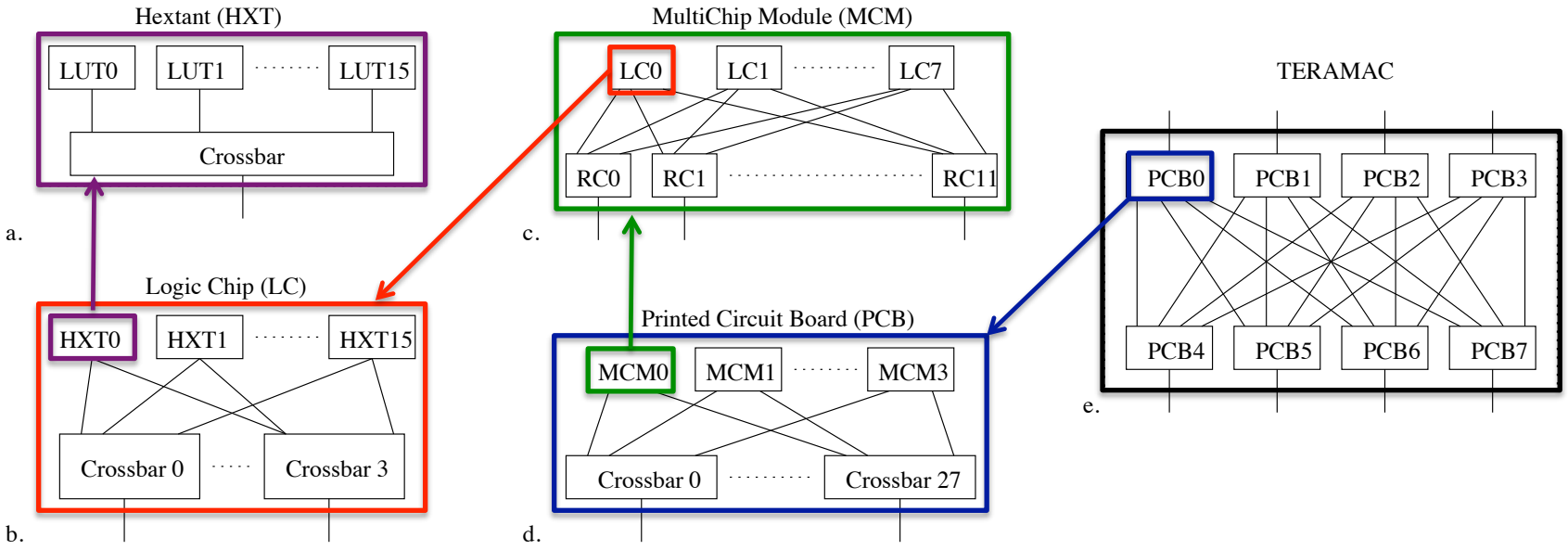
- Defectuous parts of the circuit are isolated
- Defect-free parts are used to implement the target function

Based on coding

- Very popular for memories
- Information redundancy

Teramac Design Hierarchy

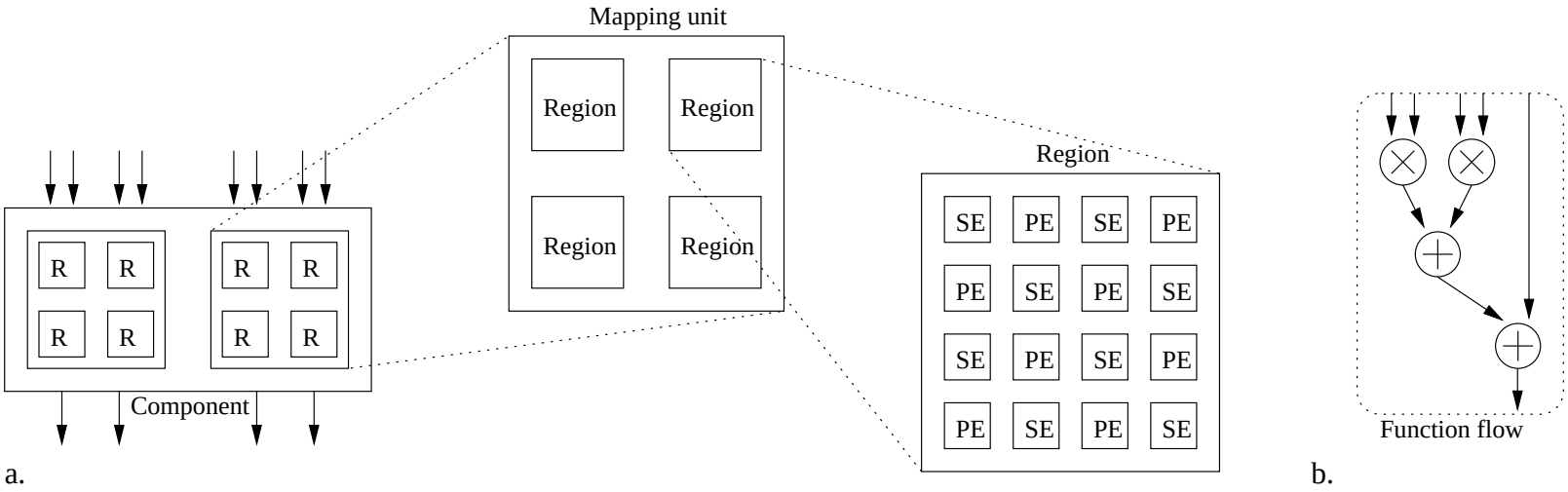
- Defect-tolerance based on wires, switches, memory



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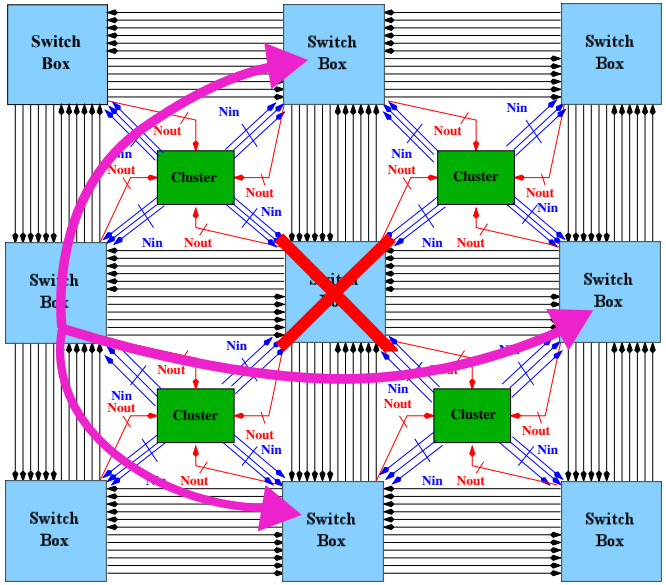
Hierarchic Nanofabric

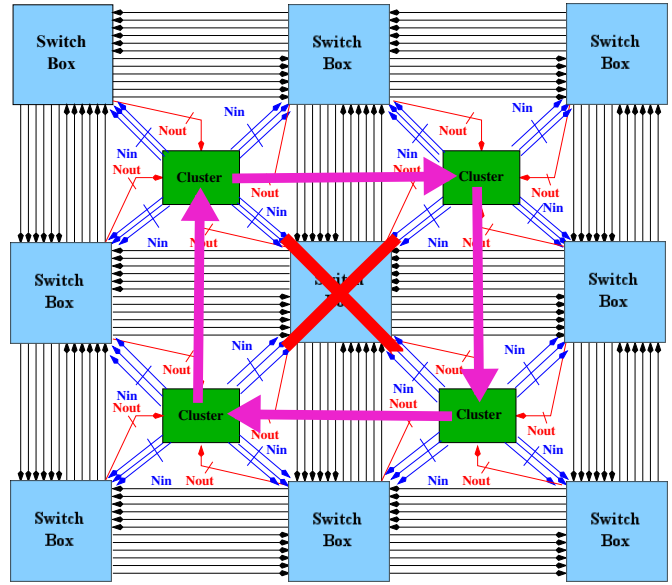
- Reconfiguration at a coarser grain
  - PEs can perform 8-bit arithmetic and logic operations
  - Tries to minimize time-consuming task of testing an mapping all of the nanofabric resources



SE= switch element, PE=processing element

FPGA: additional connections

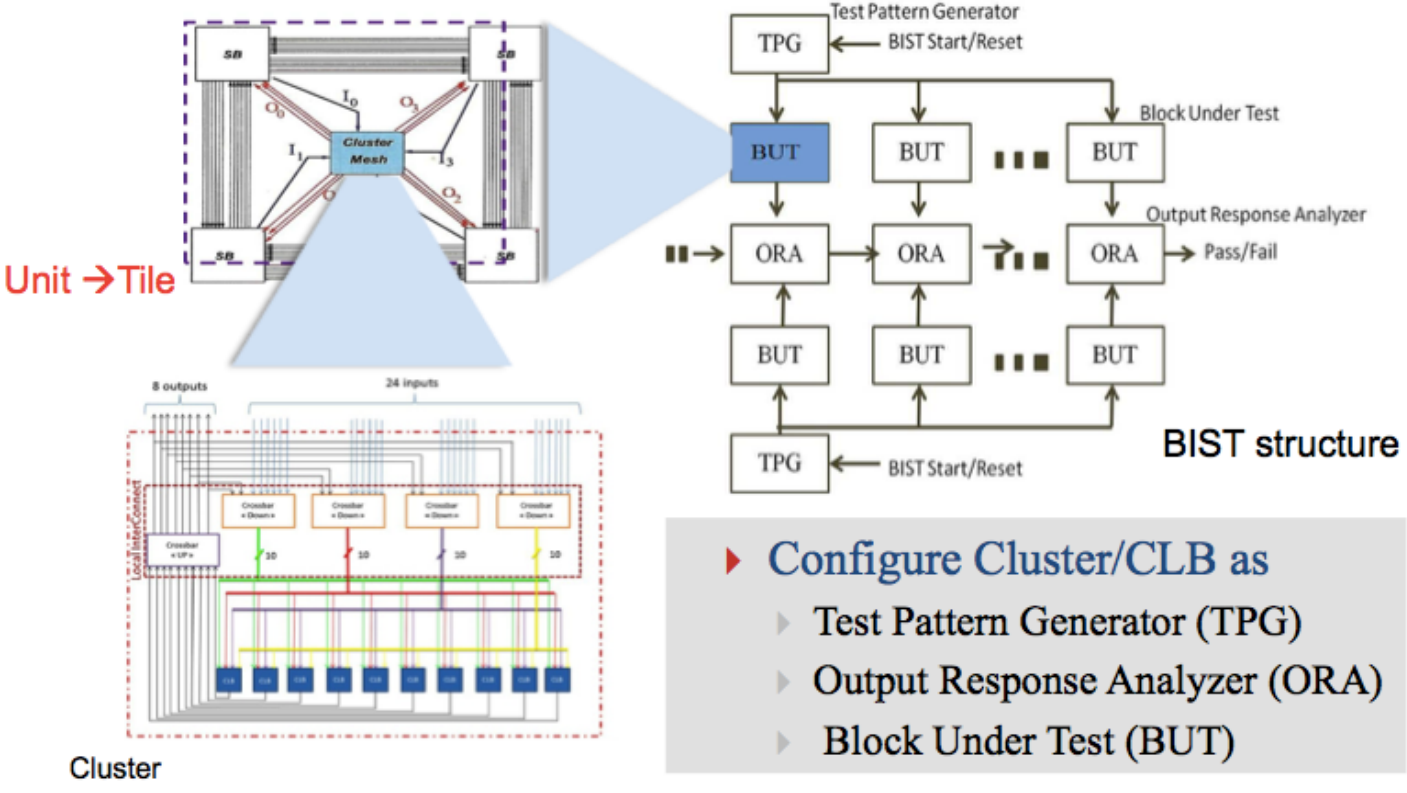




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FPGA: additional connections (cont.)

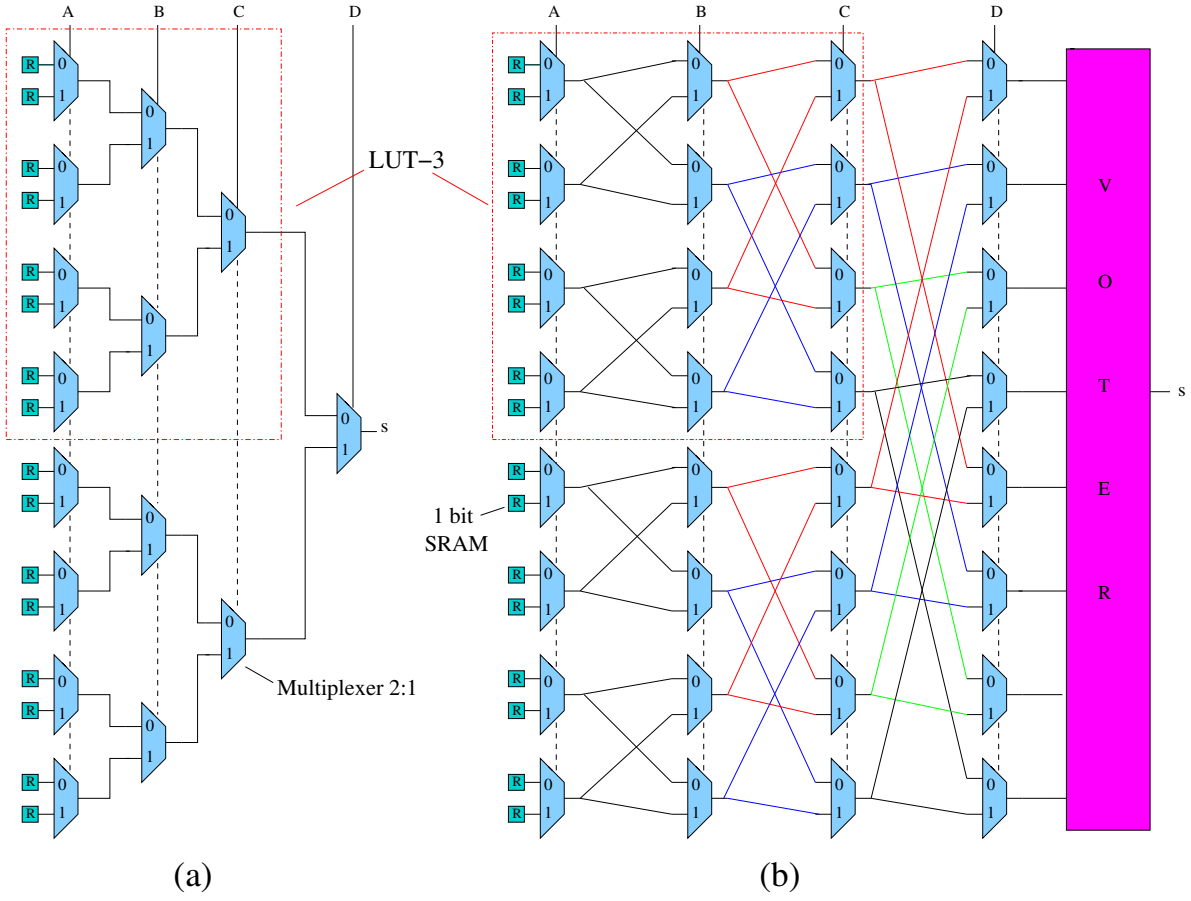
Defaults Cartography



- Configure Cluster/CLB as
  - Test Pattern Generator (TPG)
  - Output Response Analyzer (ORA)
  - Block Under Test (BUT)

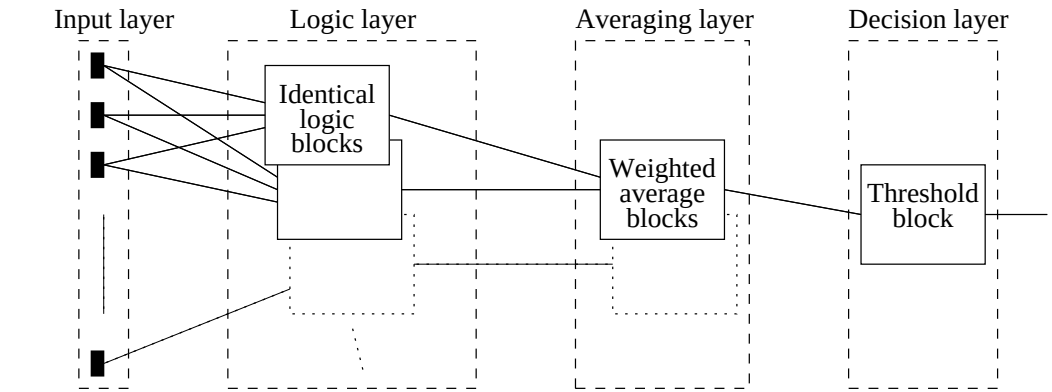
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FPGA: additional basic blocks

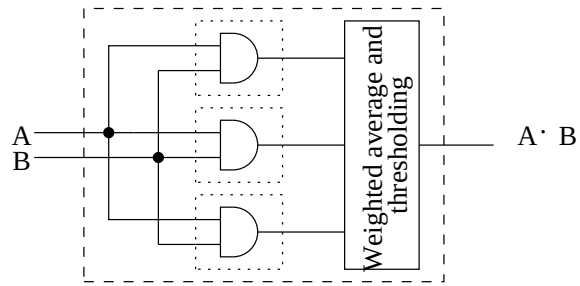


Multiple-valued Logic Approach

- Use of bit stream operators



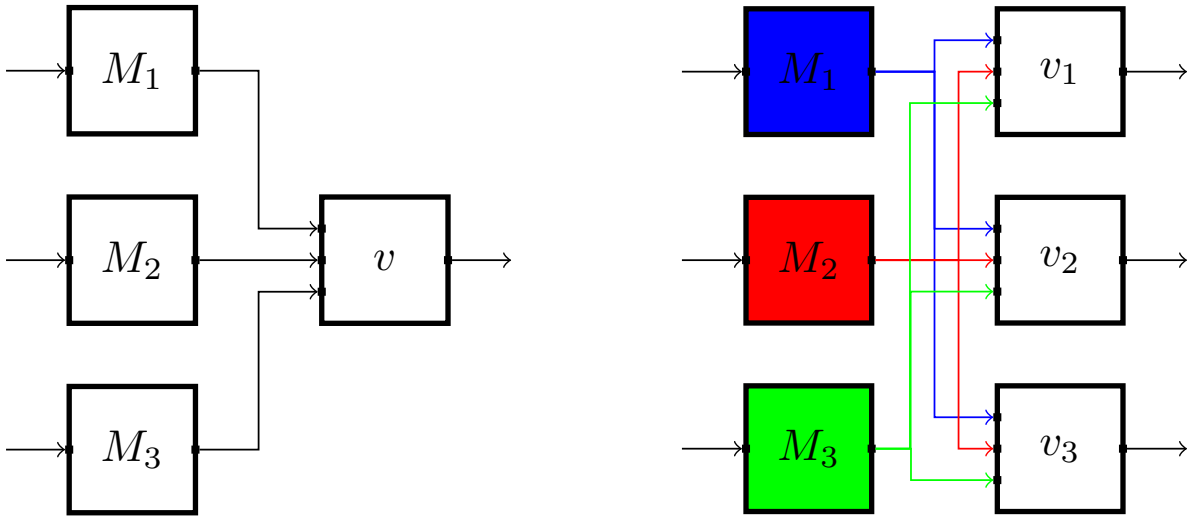
a.



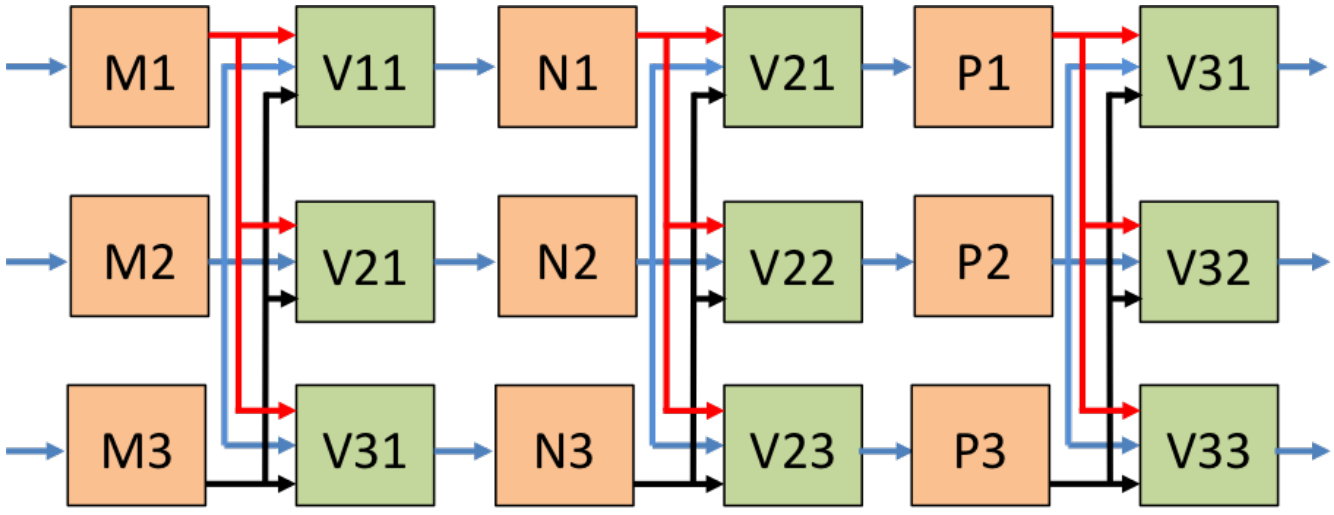
b.

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Triple Modular Redundancy

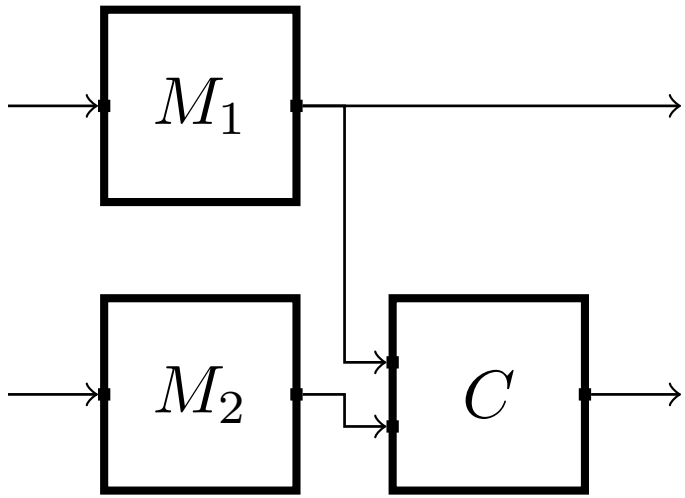


Triple Modular Redundancy (cont.)



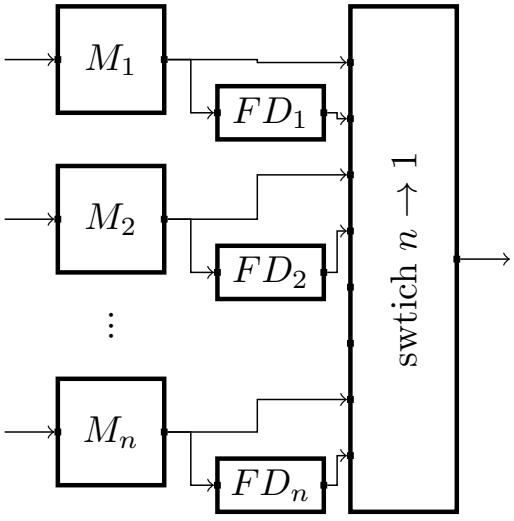
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Duplication With Comparison



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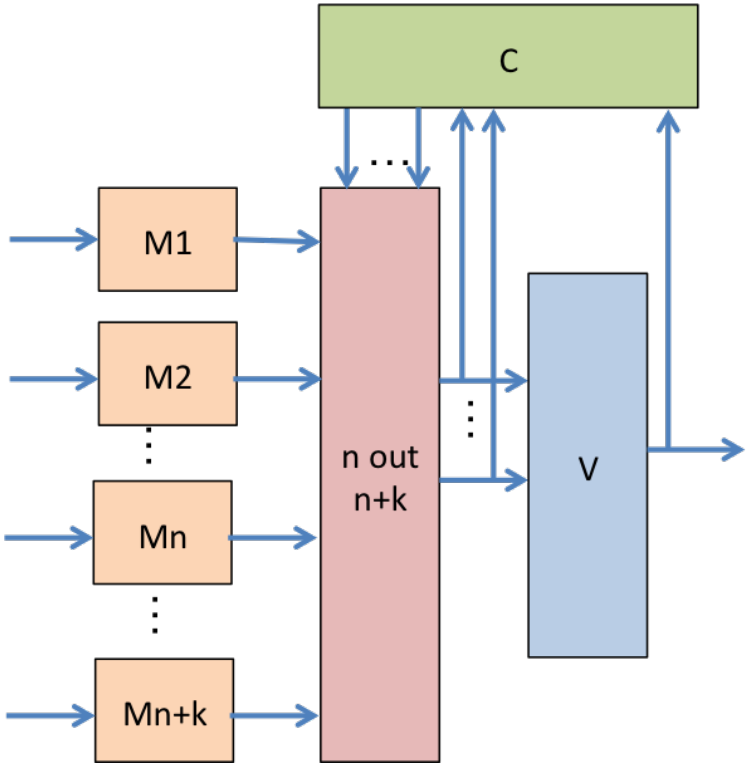
Standby Redundancy



- Only one block is active at a time.
- The **FD** blocks implement fault detection (often code based).

- Pair and Spare: similar to standby redundancy, but with two active modules

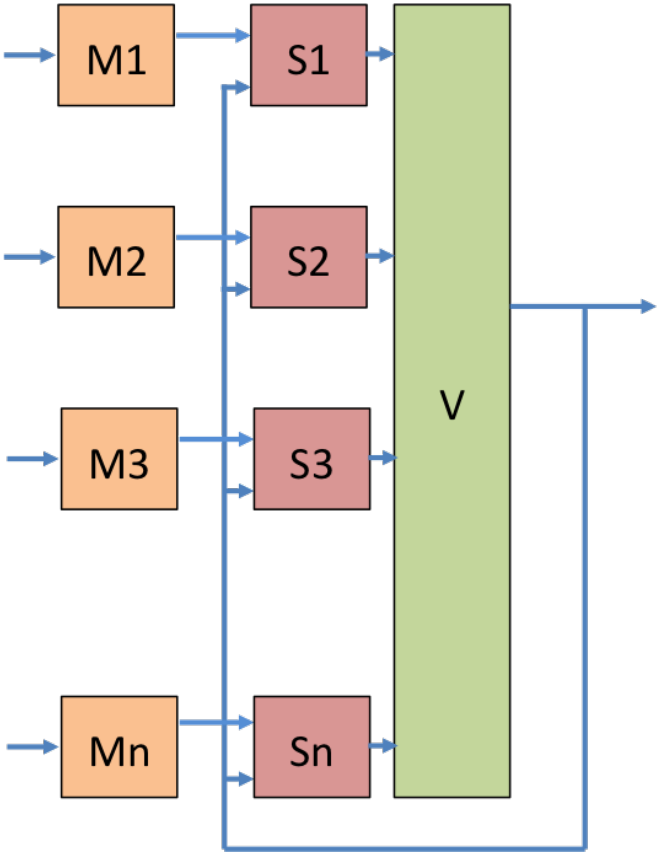
NMR with Spare



- n blocks are active at a time.
- The block **C** detects faulty blocks. It controls the action of the n-out-$n+k$ selector
- A faulty block is replaced by a spare one.

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NMR with Purging

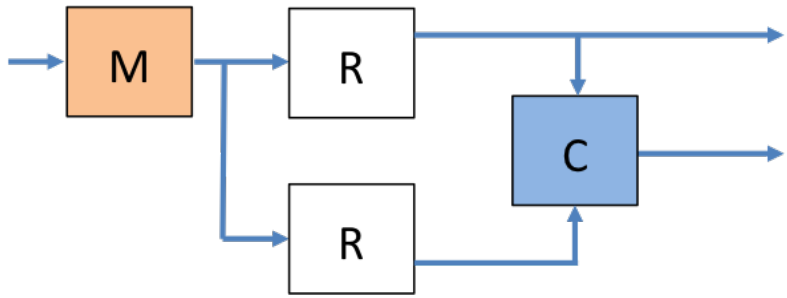


- All blocks are active in the beginning.
- The voter considers weighted inputs.
- S blocks use voter's output to change modules weights ($w = 0$ for a faulty block).

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Time Redundancy & Recomputing

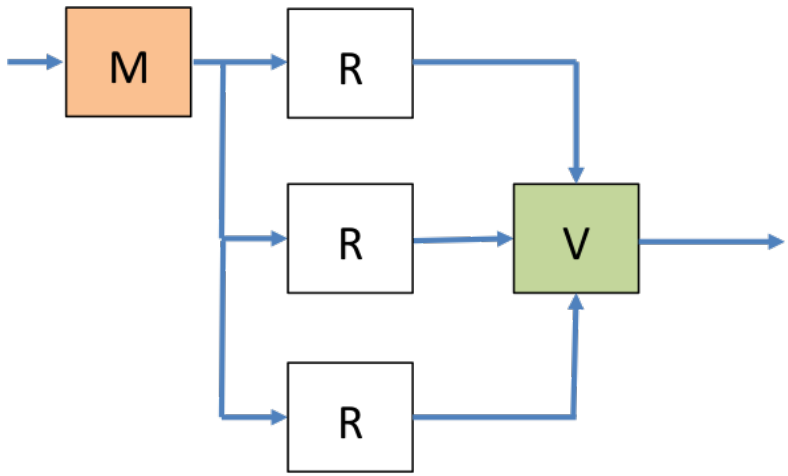
Transient faults: Duplication



Fault detection

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Transient faults: Triplication

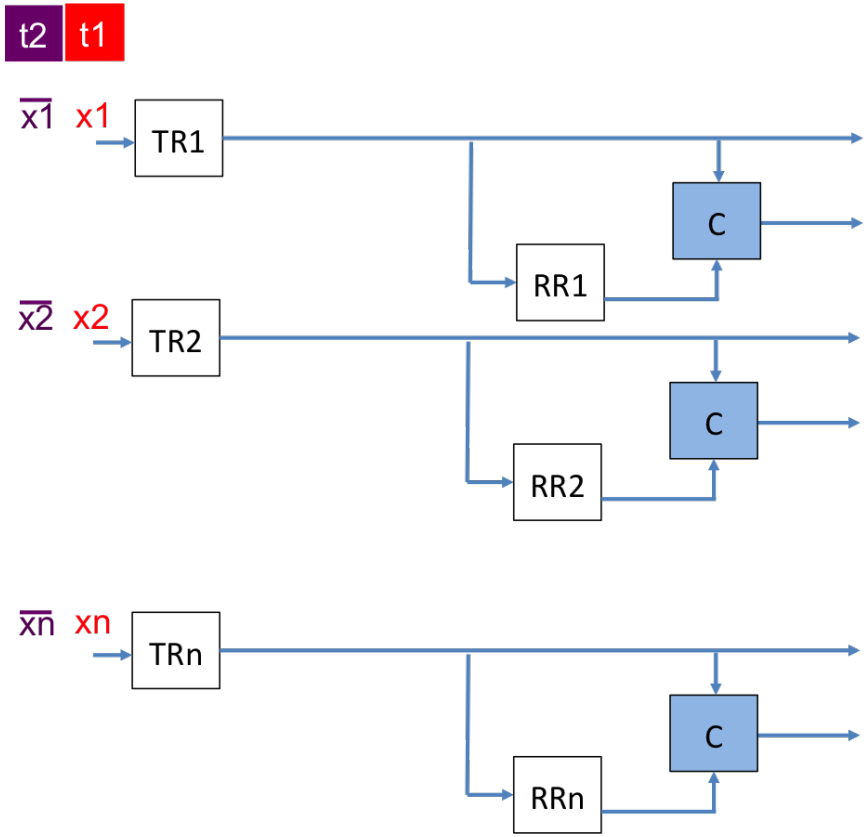


Fault masking

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Recomputing: Alternating Logic

Useful for permanent faults



Permanent fault detection

Recomputing: Operands Shifting

Useful for permanent faults

-	x_{n-1}	$\cdots$	x_{i+1}	x_i	$\cdots$	x_1	x_0
-	y_{n-1}	$\cdots$	y_{i+1}	y_i	$\cdots$	y_1	y_0
-	z_{n-1}	$\cdots$	z_{i+1}	z_i	$\cdots$	z_1	z_0

time $= k$

x_{n-1}	x_{n-2}	$\cdots$	x_i	x_{i-1}	$\cdots$	x_0	-
y_{n-1}	y_{n-2}	$\cdots$	y_i	y_{i-1}	$\cdots$	y_0	-
z_{n-1}	z_{n-2}	$\cdots$	z_i	z_{i-1}	$\cdots$	z_0	-

time $= k + 1$

c_{n-1}	c_{n-2}	$\cdots$	c_i	c_{i-1}	$\cdots$	c_0	-

check result

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Recomputing: Duplication & Comparison

- Split the inputs into two parts:
  $A = A_H \| A_L$ and $B = B_H \| B_L$
- Time t_1:
  $f(A^L, B^L)$, with $A^L = A_L \| A_L$ and $B^L = B_L \| B_L$
- Time t_2:
  $f(A^H, B^H)$, with $A^H = A_H \| A_H$ and $B^H = B_H \| B_H$

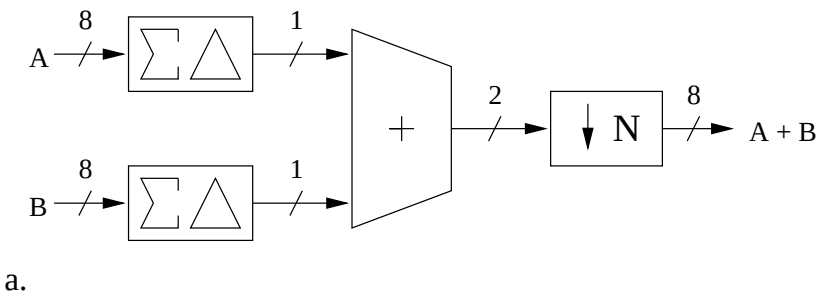
Recomputing: Swapped Operands

- Split the inputs into two parts:
  $A = A_H \| A_L$ and $B = B_H \| B_L$
- Time t_1:
  $f(A, B)$
- Time t_2:
  $f(A^s, B^s)$, with $A^s = A_L \| A_H$ and $B^s = B_L \| B_H$

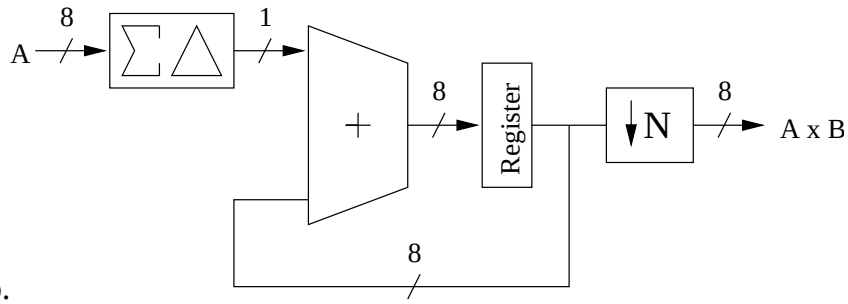
Time Redundancy

Sigma Delta Modulation Approach

- Use of bit stream operators



a.



b.

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Information Redundancy

- Add redundant bits to the information representation
- Mainly used for memories (data storage)
  - Error correcting coding (ECC)
- Hardware/time redundancy can be viewed as information redundancy

Parity Coding

- We consider a $n-$bits code
  - $k = n - 1$ bits of information
  - 1 check bit
- Example:
  - Even parity: $\vec{x} = 1010 \Rightarrow \vec{c} = 10100$
  - Odd parity: $\vec{x} = 1010 \Rightarrow \vec{c} = 10101$

x_2	x_1	x_0	c_3	c_2	c_1	c_0
0	0	0	0	0	0	0
0	0	1	0	0	1	1
0	1	0	0	1	0	1
0	1	1	0	1	1	0
1	0	0	1	0	0	1
1	0	1	1	0	1	0
1	1	0	1	1	0	0
1	1	1	1	1	1	1

Even parity coding

Berger Codes

- Check bits represent the number w of $1's$ in the data word
- Code length is $n = k + m$, where $m = \lfloor \log_2(k+1) \rfloor$
- Check bits are given by the complement of w's binary representation

x_2	x_1	x_0	c_4	c_3	c_2	c_1	c_0
0	0	0	0	0	0	1	1
0	0	1	0	0	1	1	0
0	1	0	0	1	0	1	0
0	1	1	0	1	1	0	1
1	0	0	1	0	0	1	0
1	0	1	1	0	1	0	1
1	1	0	1	1	0	0	1
1	1	1	1	1	1	0	0

$\vec{x} = 0010 \Rightarrow \vec{c} = 0010\mathbf{110}$

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(n, k) Linear Code

- Defined over the finite field $\mathbb{F}_q$
  - $\mathbb{F}_2$ is the set $\{0, 1\}$ with $\oplus$ and $\cdot$ operations
- k-dimensional subspace $\mathbb{C}_k$ of a vector space $\mathbb{V}_n$
  - $\mathbb{V}_n$ is a subset of $\mathbb{F}_q^n$ with with addition and multiplication by scalar operations
    $\mathbb{F}_2^n$ is the set of all n-uples containing elements of $\mathbb{F}_2$.
- $\mathbb{C}_k$ subspace is based on k linearly independent vectors.
  - Any codeword of a (n, k) linear code can be written as a linear combination of the k basis vectors $\{\vec{v}_0, \vec{v}_1 ..., \vec{v}_{k-1}\}$ of subspace $\mathbb{C}_k$

$$\vec{c} = \sum_{j=0}^{k-1} \vec{x}_i \vec{v}_i$$

Matrix Representation

$$\vec{c} = \vec{x} \cdot \mathbf{G}$$

- $\vec{c}$ is the codeword
- $\vec{x}$ is the information word
- $\mathbf{G}$ generator matrix
  - The rows of $\mathbf{G}$ are k vectors which are a basis of C.
  - $\mathbf{G}$ has n columns.

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Example

- Code (5,3) over $\mathbb{F}_2$
- Generator matrix

$$\mathbf{G} = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{pmatrix}$$

x_2	x_1	x_0	c_4	c_3	c_2	c_1	c_0
0	0	0	0	0	0	0	0
0	0	1	0	0	1	1	1
0	1	0	0	1	0	0	1
0	1	1	0	1	1	1	0
1	0	0	1	0	0	1	0
1	0	1	1	0	1	0	1
1	1	0	1	1	0	1	1
1	1	1	1	1	1	0	0

$$\begin{pmatrix} x_2 & x_1 & x_0 \end{pmatrix} \cdot \mathbf{G} = \begin{pmatrix} c_4 & c_3 & c_2 & c_1 & c_0 \end{pmatrix}$$

$$\begin{pmatrix} c_4 & c_3 & c_2 & c_1 & c_0 \end{pmatrix} = \begin{pmatrix} x_2 & x_1 & x_0 & x_2 \oplus x_0 & x_1 \oplus x_0 \end{pmatrix}$$

Parity Check Matrix H

- $\mathbf{H}$ is a $(n-k) \times k$ matrix used to detect errors

$$\mathbf{H}.\mathbf{G}^T = 0$$

- The product of $\mathbf{H}$ by a vector produces a syndrome vector

$$\mathbf{H}.\vec{r}^T = \vec{s} \text{ where } \vec{s} = \vec{0} \text{ if } \vec{c} \text{ is a codeword}$$

- If $\mathbf{G} = [\mathbf{I}_k \mathbf{A}]$, then $\mathbf{H} = \left[\mathbf{A}^T \mathbf{I}_{n-k}\right]$

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Hamming Code

$$\vec{c} = \vec{x}\mathbf{G} = (1010) \left(\begin{array}{ccc|cccc} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 \end{array}\right) = (1011010)$$

Code (7,4):
generation matrix **G**,
syndrome matrix **H**

No error $\Rightarrow \vec{s} = \vec{c}\mathbf{H}^{\mathbf{T}} = \vec{0}$

$$\vec{s} = \vec{c}\mathbf{H}^{\mathbf{T}} = (1011010) \left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \hline 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{array}\right) = (000)$$

H is in a lexicographic form

Error Vector	Syndrome $\mathbf{s} = \mathbf{cH}^{\mathbf{T}}$
1000000	100
0100000	010
0010000	001
0001000	110
0000100	101
0000010	011
0000001	111

Hamming Code

$$\vec{c} = \vec{x}\mathbf{G} = (1010) \left(\begin{array}{ccc|cccc} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 \end{array}\right) = (1011010)$$

Code (7,4):
generation matrix **G**,
syndrome matrix **H**

No error $\Rightarrow \vec{s} = \vec{c}\mathbf{H}^{\mathbf{T}} = \vec{0}$

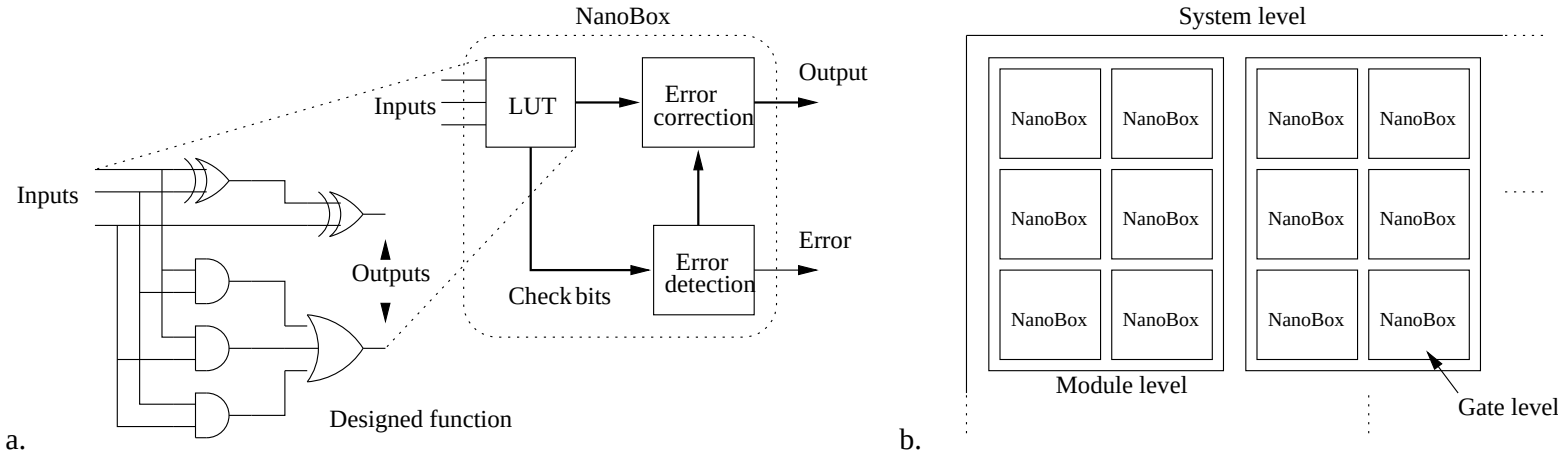
$$\vec{s} = \vec{c}\mathbf{H}^{\mathbf{T}} = (1011110) \left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \hline 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{array}\right) = (101)$$

H is in a lexicographic form

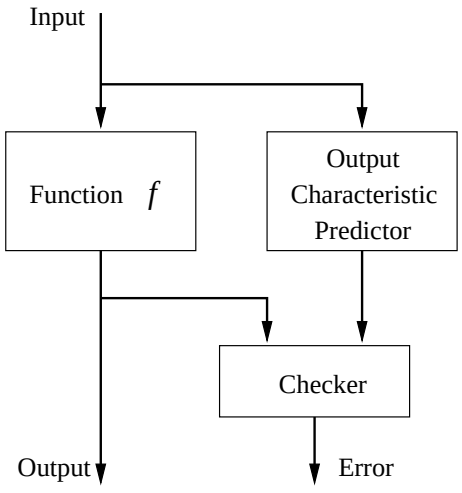
Error Vector	Syndrome $\mathbf{s} = \mathbf{cH}^{\mathbf{T}}$
1000000	100
0100000	010
0010000	001
0001000	110
0000100	101
0000010	011
0000001	111

NanoBox Approach

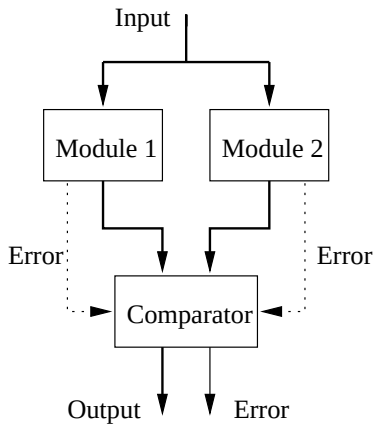
- Dense regular structures with reconfigurable capabilities



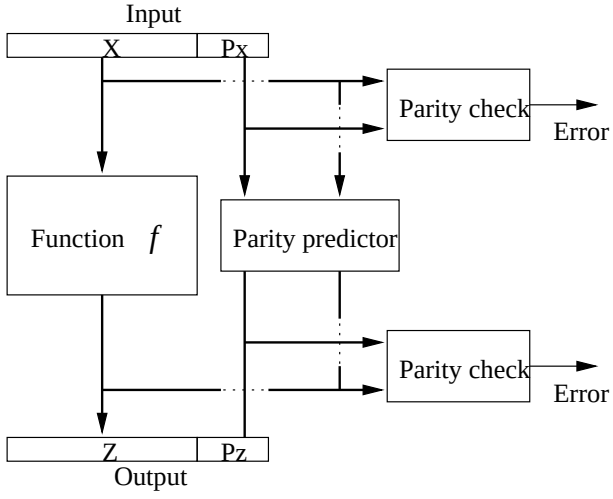
Concurrent Error Detection & Coding



a.



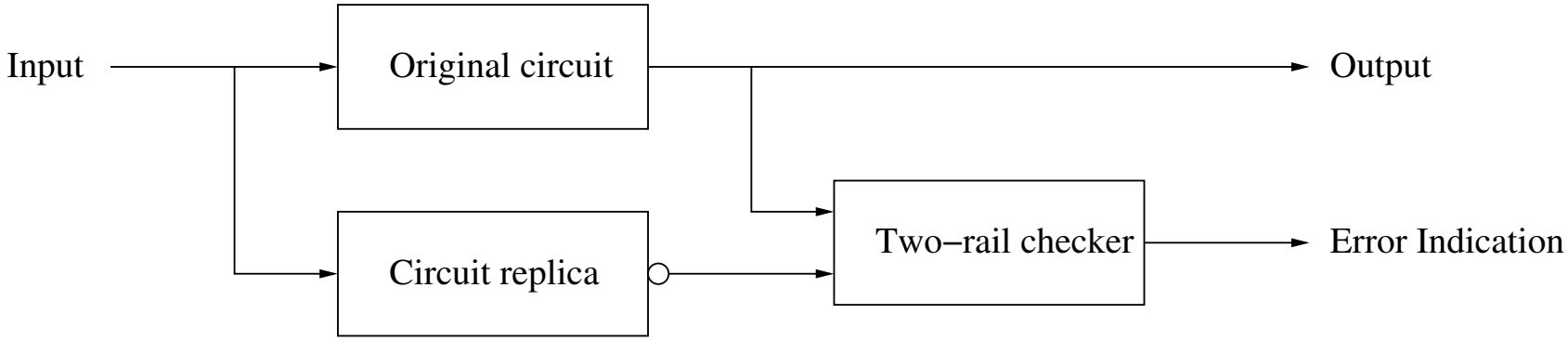
b.



c.

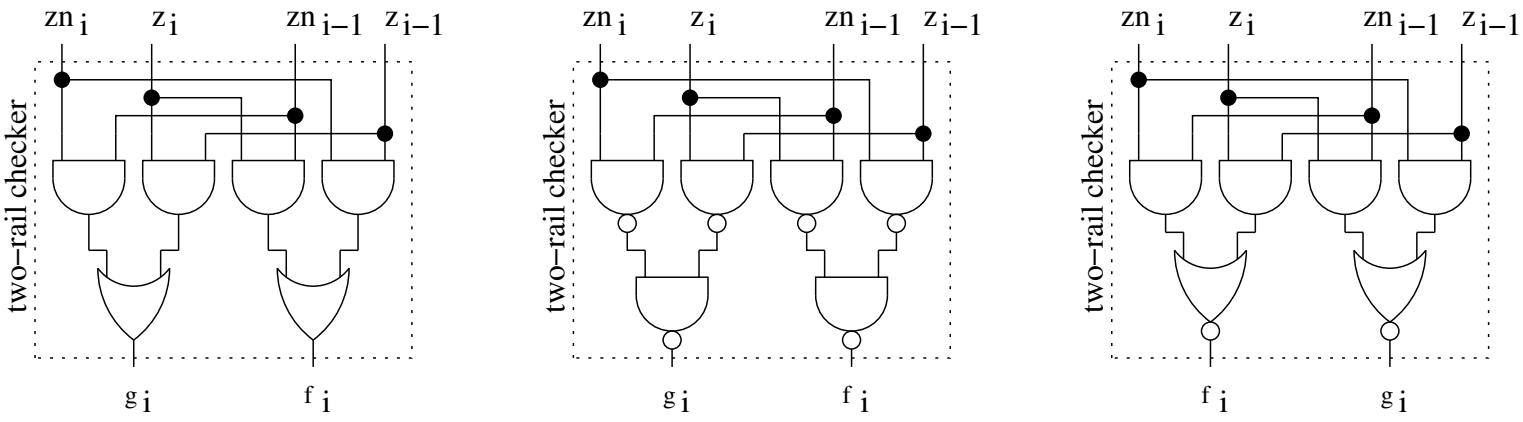
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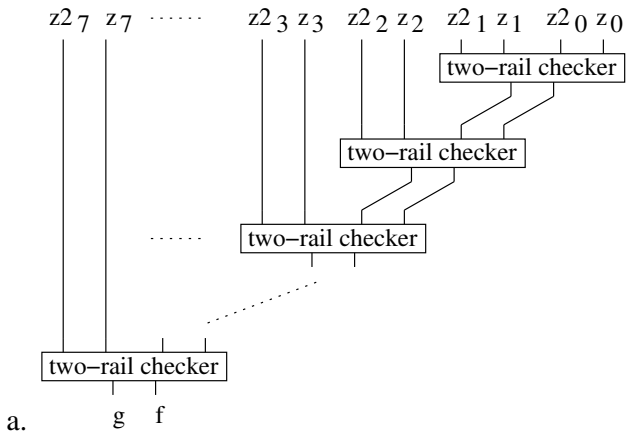
Repetition Coding



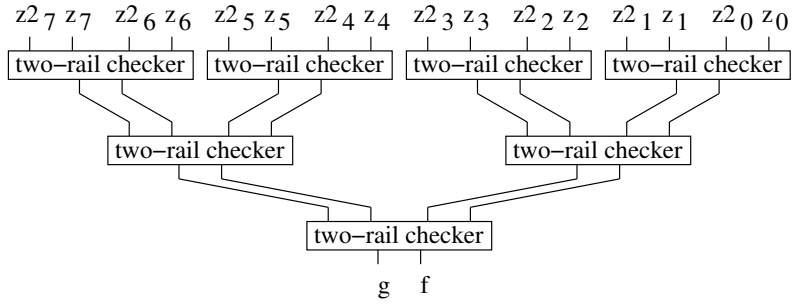
- Self-dual functions: $f(\overline{x_0}, \overline{x_1}, \cdots, \overline{x_n}) = \overline{f}(x_0, x_n, \cdots, x_n)$

Checker Implementations





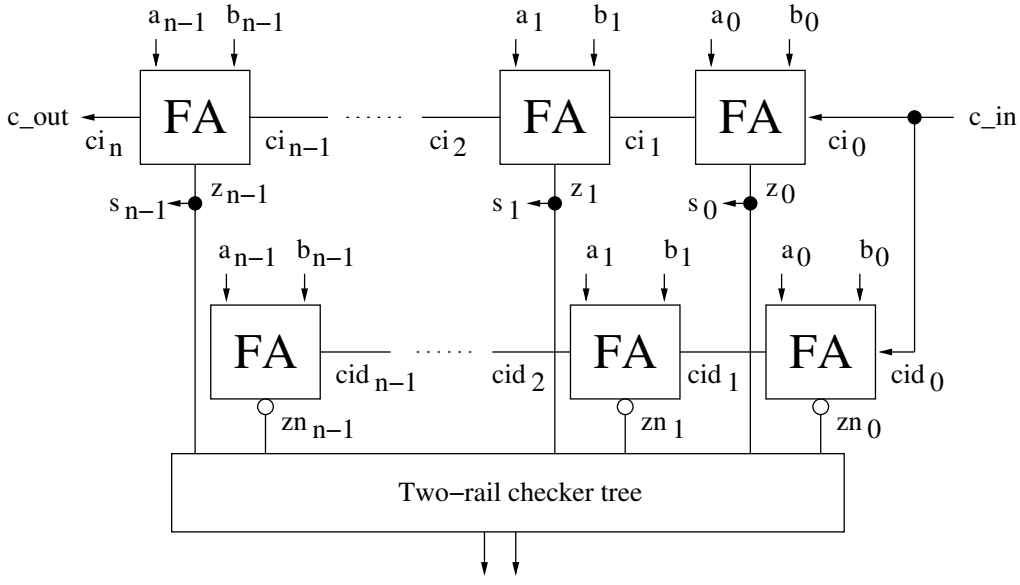
a.



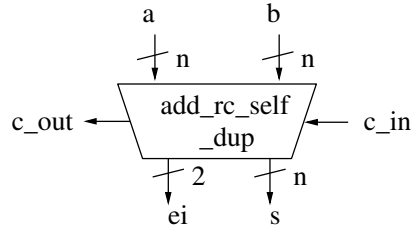
b.

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Duplicated Ripple Carry Adder



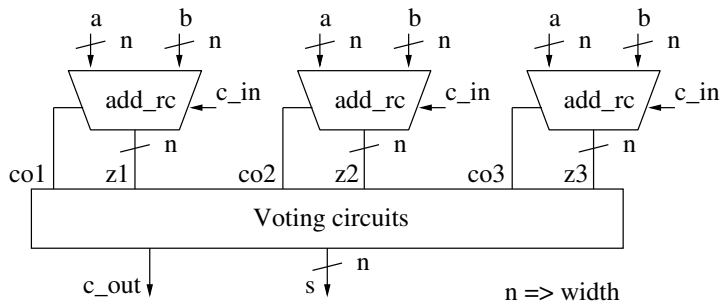
a.



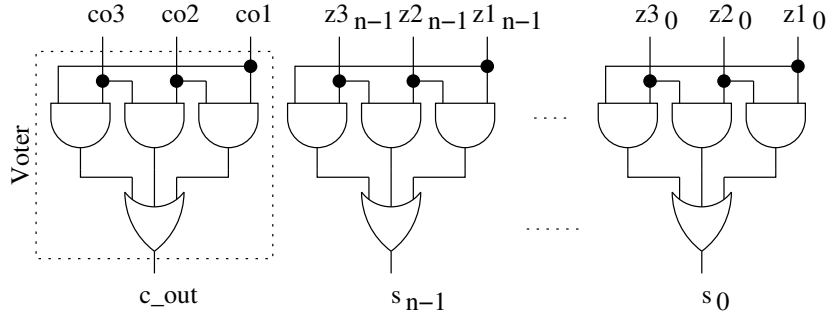
b.

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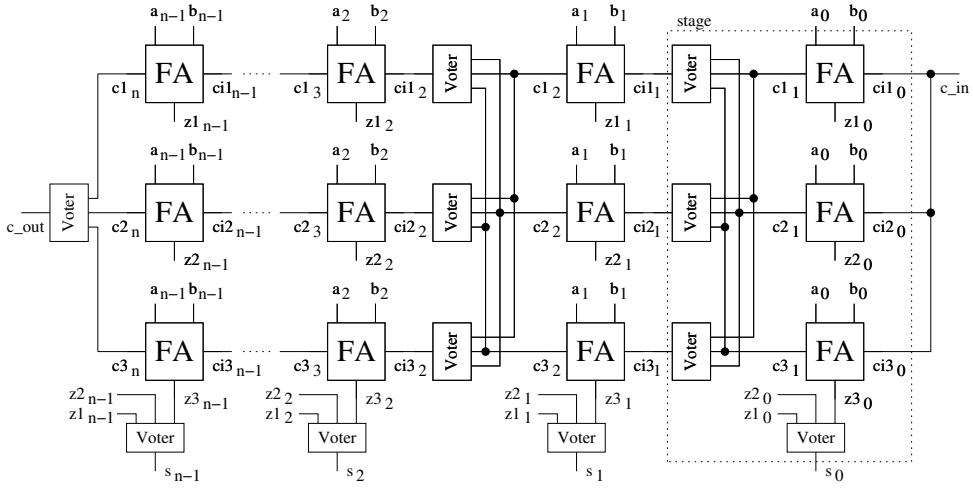
TMR Ripple Carry Adder



a.



b.



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Conclusions

- This lesson dealt with different methods for reliability improvement
  - Detection and correction of faults
  - Solutions for transient and permanent faults
  - Strategies based on passive and active redundancy
- Reliability improvement leads to area and/or time penalties
- Next slides will explore techniques for reliability assessment

Outline

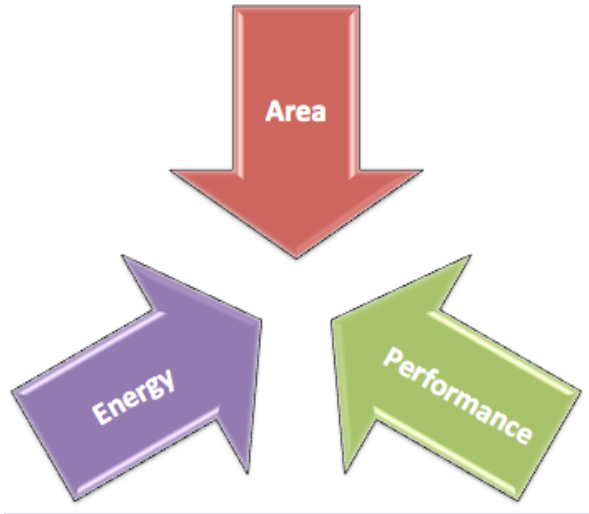
Dependability
- Introduction
- Deterministic model
- Probabilistic models

Fault and defect tolerance improvement

Fault tolerance assessment

Optimized Design of Processors





Optimized Design of Processors

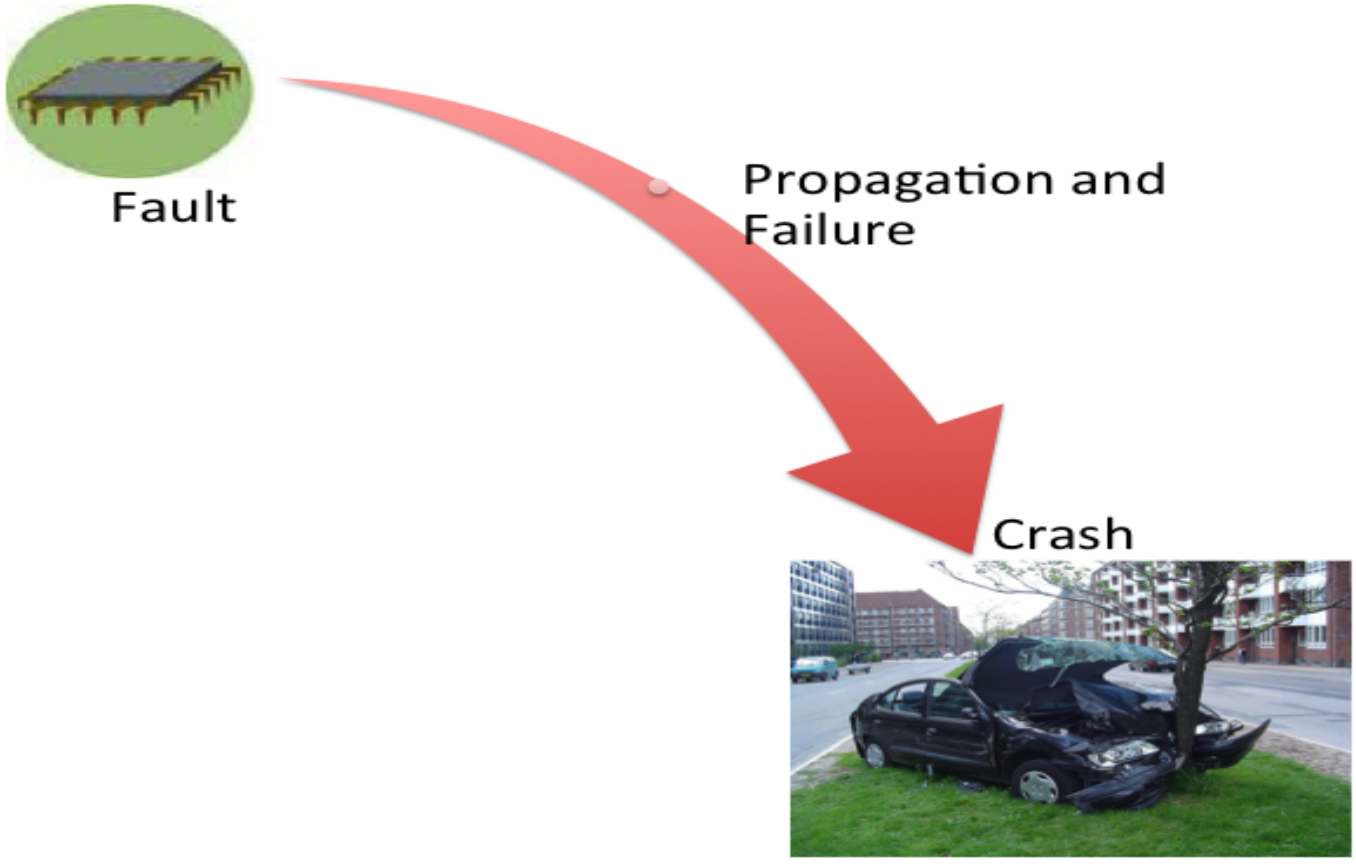
How to get **optimized** design of **reliable** processors based on **unreliable** devices?



- Risk minimization
- More (than) Moore
- Fabless generalization

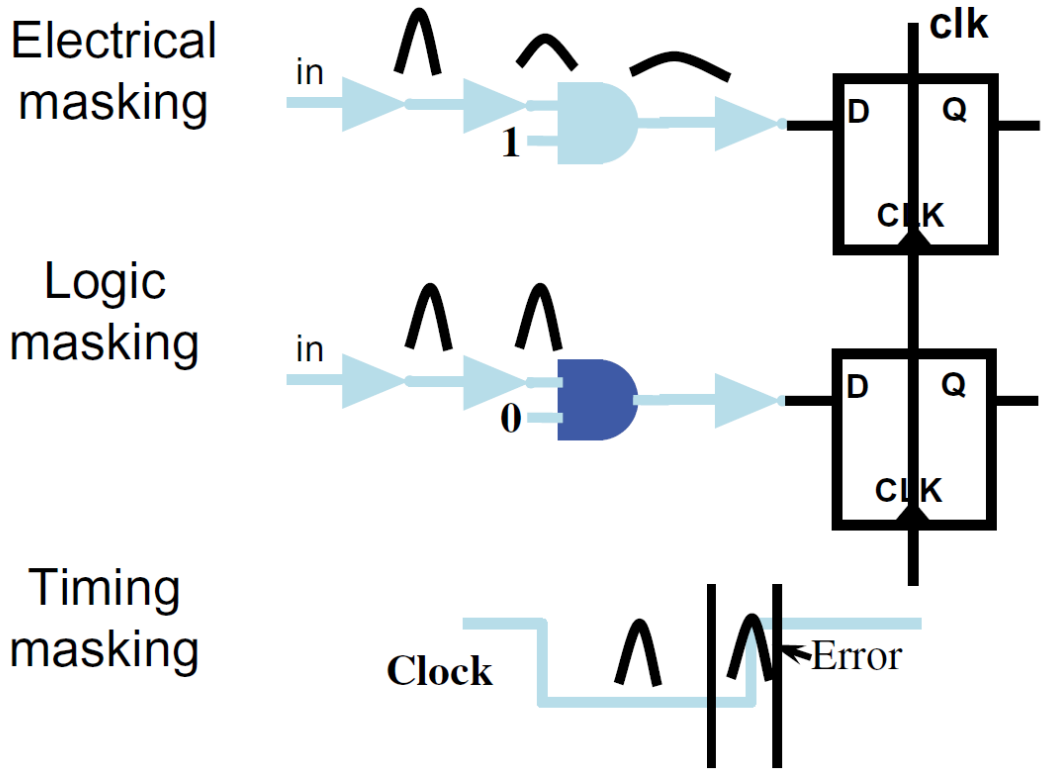
Reliability improvement
$\Rightarrow$ penalties !

Fault Propagation



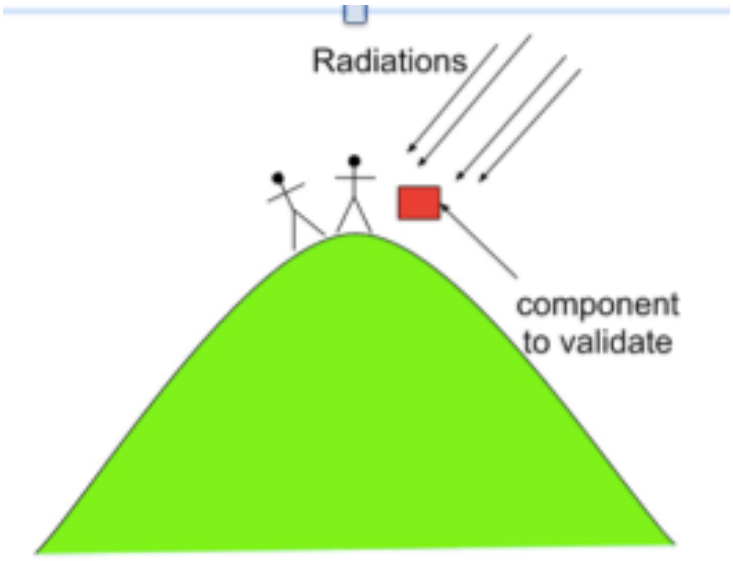
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Masking Effects



$\Rightarrow$ Reliability assessment !

Hardware Fault Injection

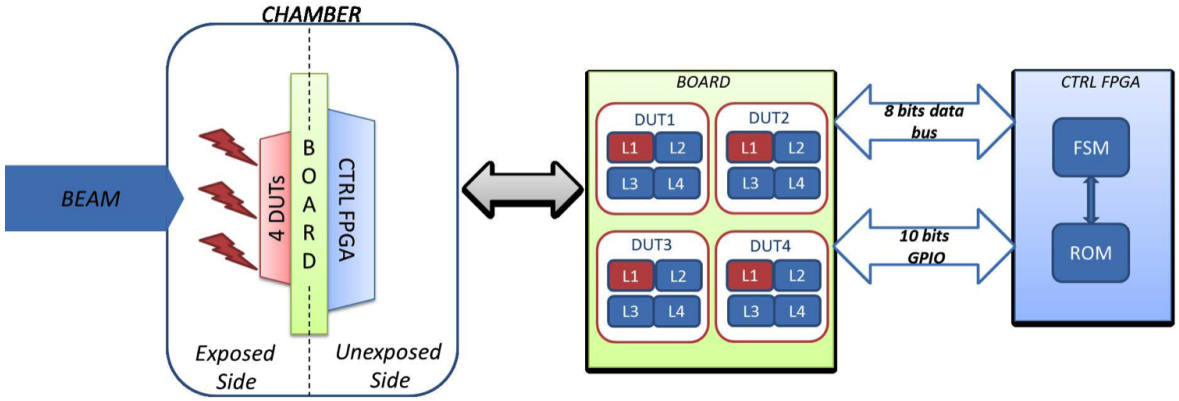


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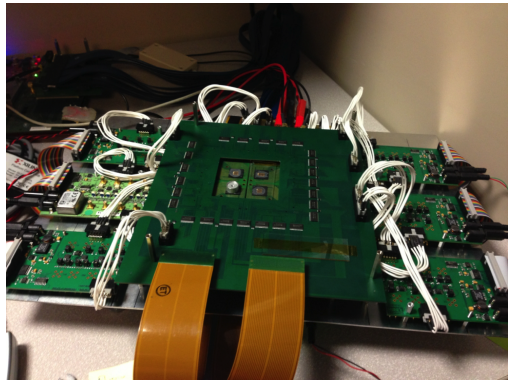
Laser Fault Injection



Heavy Ions Test on a μProc



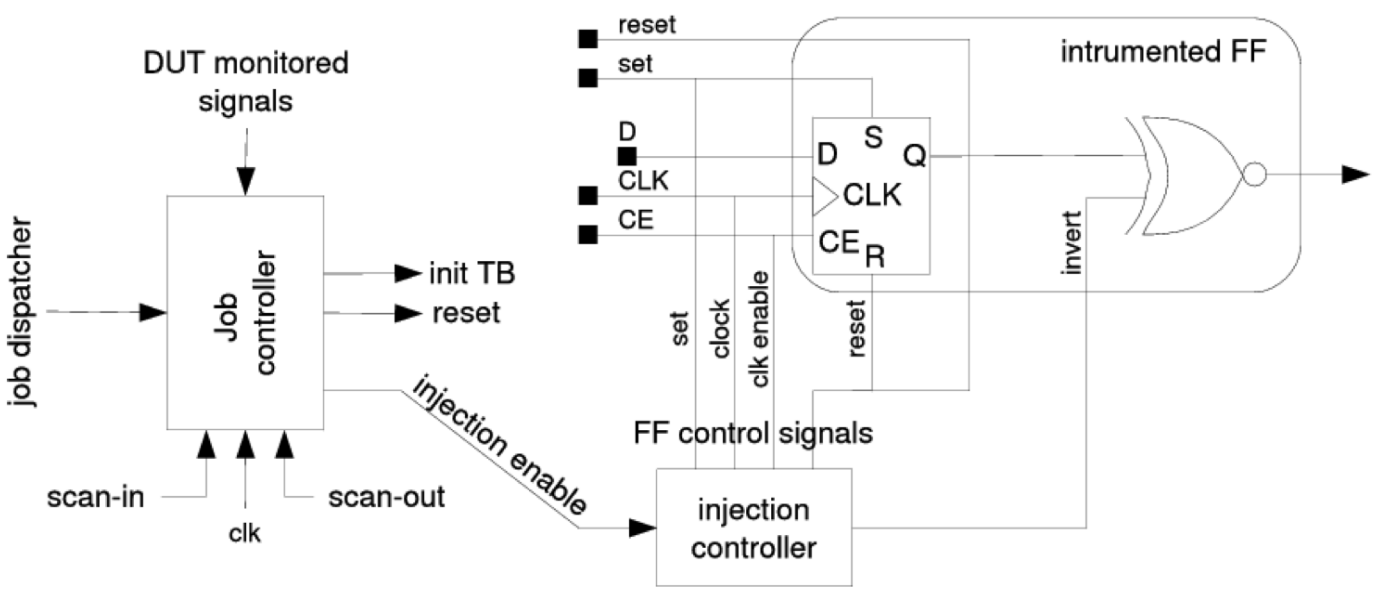
Test Board and Experimental Setup



Alpha-particles irradiation for setup validation

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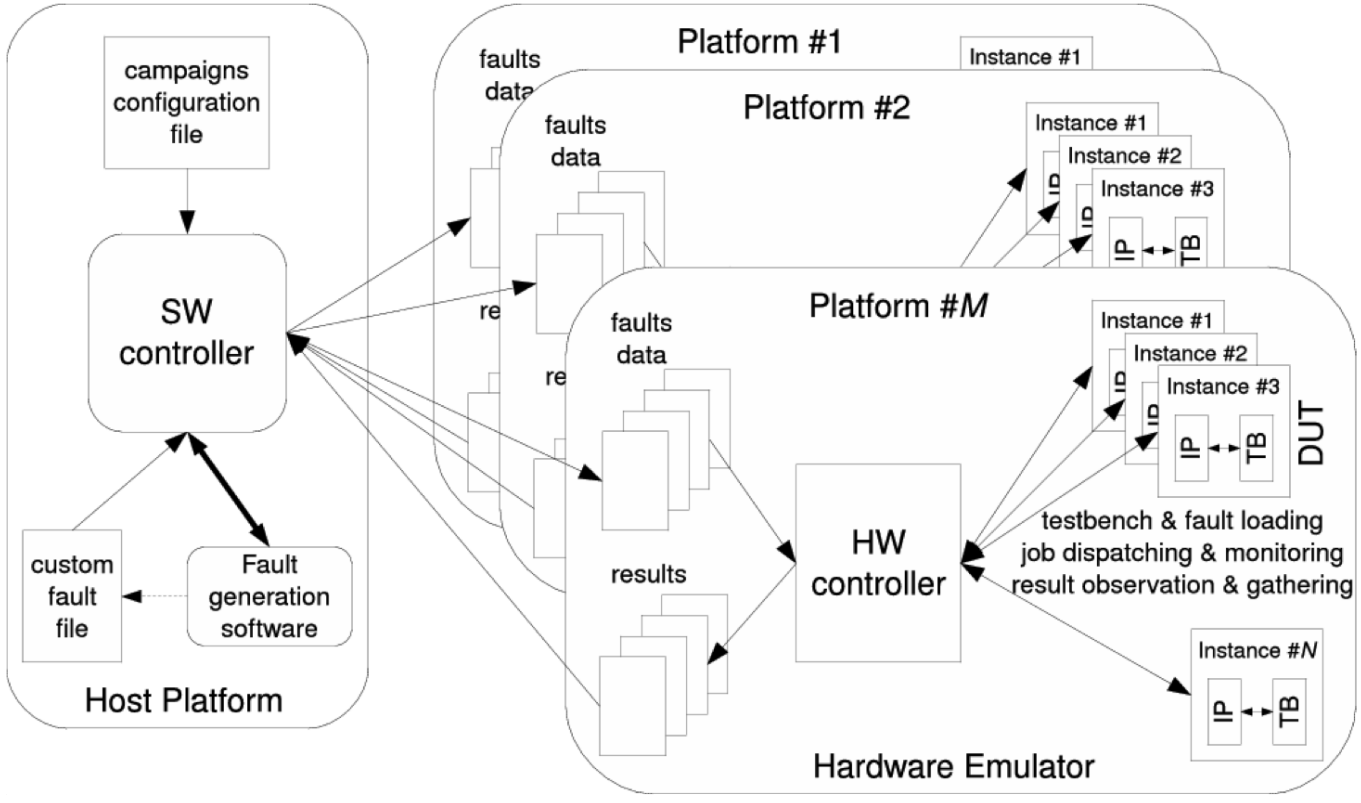
Fault Injection: HW Emulation



Saboteur. Setup for SEU Fault Injection

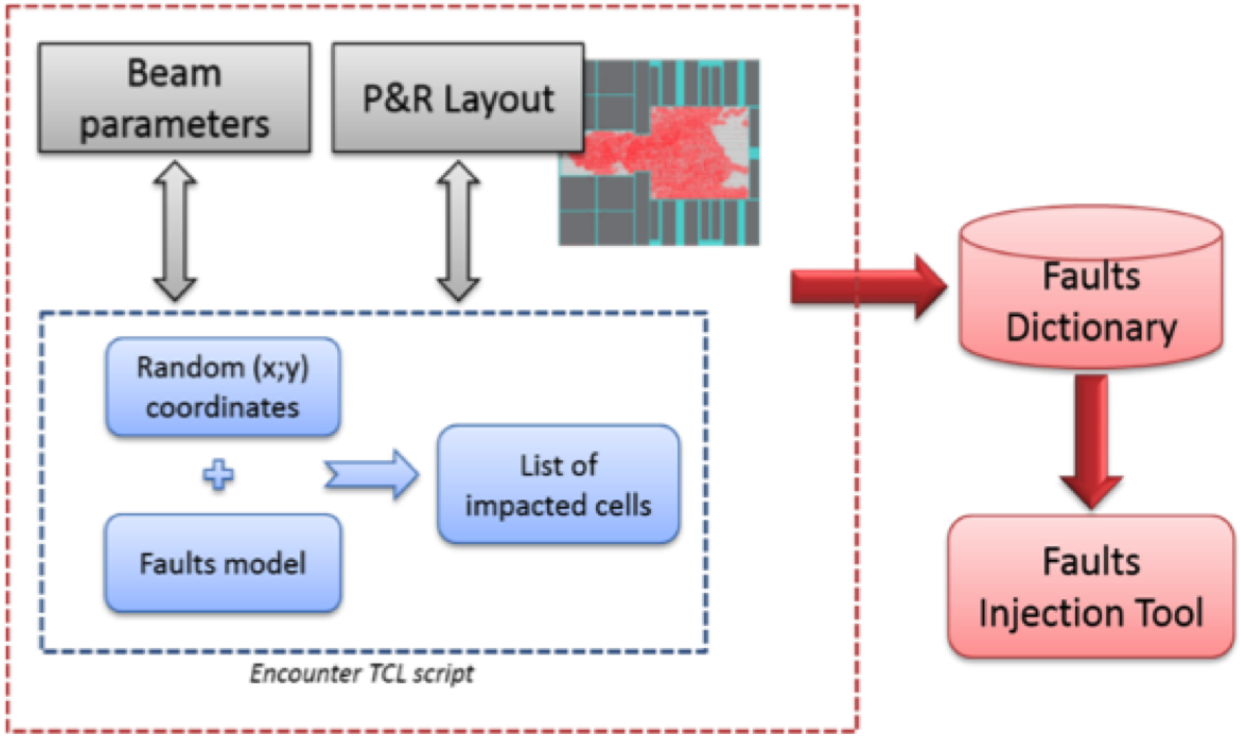
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Fault Injection: HW Emulation (cont.)



Fault Injection/Emulation Platform.

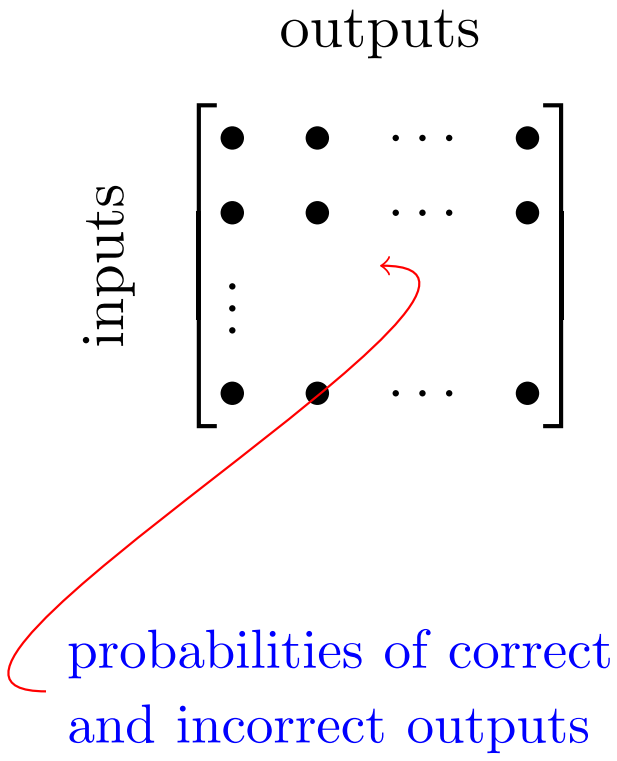
Fault Injection: Simulation



Platform General Structure

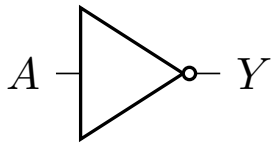
Probabilistic Transfer Matrix (PTM)

- A matrix that models the probability of output values according to the occurrence of input values and gate reliability.
- A fault-free gate is modeled by ideal transfer matrix (ITM).



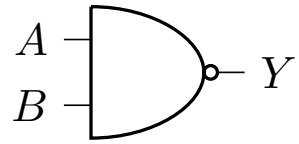
Examples

Logic gate NOT



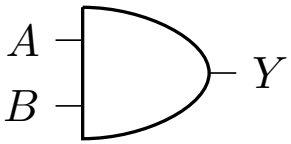
$$PTM = \begin{bmatrix} p_0 & q_0 \\ q_1 & p_1 \end{bmatrix} \qquad ITM = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Logic gate NAND



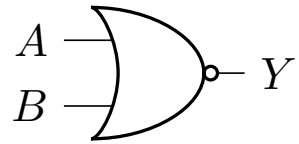
$$PTM = \begin{bmatrix} p_{00} & q_{00} \\ p_{01} & q_{01} \\ p_{10} & q_{10} \\ q_{11} & p_{11} \end{bmatrix} \qquad ITM = \begin{bmatrix} 0 & 1 \\ 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Logic gate AND



$$PTM = \begin{bmatrix} q_{00} & p_{00} \\ q_{01} & p_{01} \\ q_{10} & p_{10} \\ p_{11} & q_{11} \end{bmatrix} \qquad ITM \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Logic gate NOR



$$PTM = \begin{bmatrix} p_{00} & q_{00} \\ q_{01} & p_{01} \\ q_{10} & p_{10} \\ q_{11} & p_{11} \end{bmatrix} \qquad ITM \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}$$

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Reliability $\leftrightarrow$ PTM

Definition

$$R = \sum_{(i,j)|ITM(i,j)=1} p(j|i)p(i) = \sum_{(i,j)|ITM(i,j)=1} PTM(i,j)p(i) \quad (1)$$

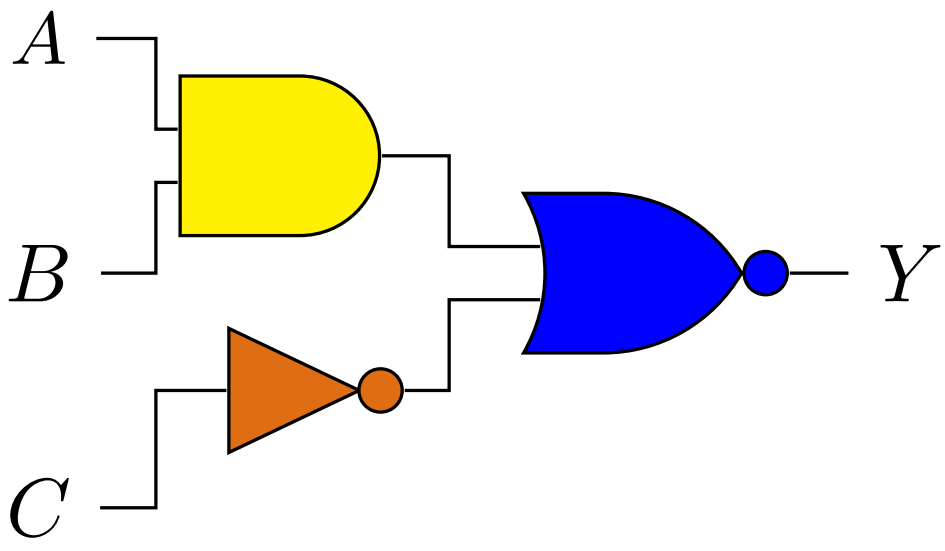
Logic gate NAND

$$R = P\{00,1\} + P\{01,1\} + P\{10,1\} + P\{11,0\}$$

$$P_{AB} = \begin{bmatrix} a_0b_0 \\ a_0b_1 \\ a_1b_0 \\ a_1b_1 \end{bmatrix} \quad PTM_{NAND} = \begin{bmatrix} p_{00} & q_{00} \\ p_{01} & q_{01} \\ p_{10} & q_{10} \\ q_{11} & p_{11} \end{bmatrix} \quad ITM_{NAND} \begin{bmatrix} 0 & 1 \\ 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$R = a_0b_0q_{00} + a_0b_1q_{01} + a_1b_0q_{10} + a_1b_1q_{11}$$

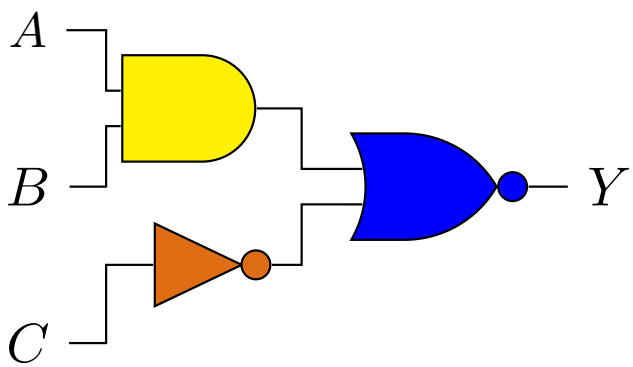
Example



$$p_{ij} = p \text{ and } q_{ij} = q \text{ for all } (i, j)$$

Parallel blocks: Kronecker Product

$$M_1 = \begin{bmatrix} q_{00} & p_{00} \\ q_{01} & p_{01} \\ q_{10} & p_{10} \\ p_{11} & q_{11} \end{bmatrix} \quad M_2 = \begin{bmatrix} p_0 & q_0 \\ q_1 & p_1 \end{bmatrix} \qquad M_{L_1} = M_1 \otimes M_2$$



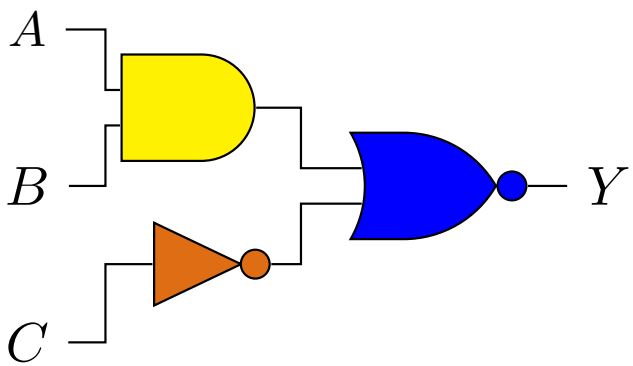
$$\mathrm{M}_{L_1} = \begin{bmatrix} pq & q^2 & p^2 & pq \\ q^2 & pq & pq & p^2 \\ pq & q^2 & p^2 & pq \\ q^2 & pq & pq & p^2 \\ pq & q^2 & p^2 & pq \\ q^2 & pq & pq & p^2 \\ p^2 & pq & pq & q^2 \\ pq & p^2 & q^2 & pq \end{bmatrix}$$

$p_{ij} = p$ and $q_{ij} = q \forall (i,j)$

Series blocks: Scalar Product

$$M_3 = \begin{bmatrix} p_{00} & q_{00} \\ q_{01} & p_{01} \\ q_{10} & p_{10} \\ q_{11} & p_{11} \end{bmatrix}$$

$$PTM_{circ} = M_{circ} = M_{L1}.M_3$$



$$\begin{bmatrix} 2p^2q + pq^2 + q^3 & p^3 + p^2q + 2pq^2 \\ p^2q + 3pq^2 & p^3 + 2p^2q + q^3 \\ 2p^2q + pq^2 + q^3 & p^3 + p^2q + 2pq^2 \\ p^2q + 3pq^2 & p^3 + 2p^2q + q^3 \\ 2p^2q + pq^2 + q^3 & p^3 + p^2q + 2pq^2 \\ p^2q + 3pq^2 & p^3 + 2p^2q + q^3 \\ p^3 + 2pq^2 + q^3 & 3p^2q + pq^2 \\ 2p^2q + pq^2 + q^3 & p^3 + p^2q + 2 \end{bmatrix}$$

$p_{ij} = p$ and $q_{ij} = q \forall (i,j)$

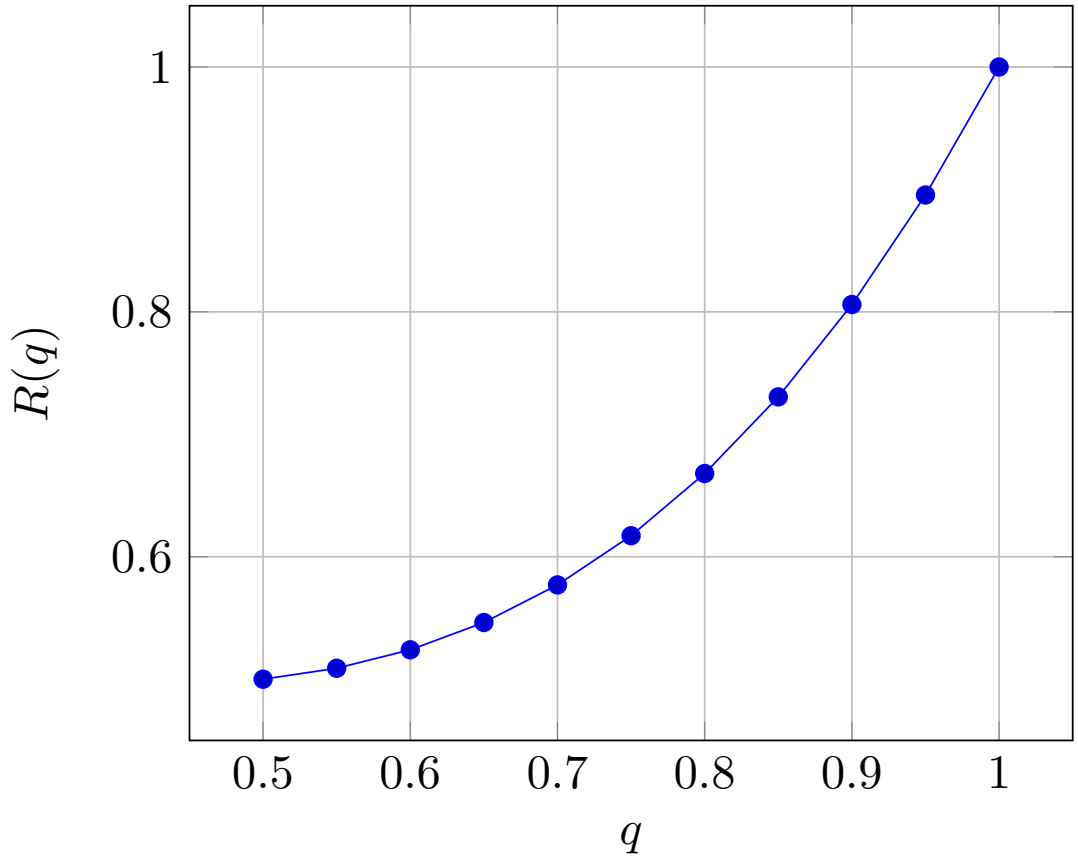
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Reliability Calculation

$$M_{circ} = \begin{bmatrix} 2p^2q + pq^2 + q^3 & p^3 + p^2q + 2pq^2 \\ p^2q + 3pq^2 & p^3 + 2p^2q + q^3 \\ 2p^2q + pq^2 + q^3 & p^3 + p^2q + 2pq^2 \\ p^2q + 3pq^2 & p^3 + 2p^2q + q^3 \\ 2p^2q + pq^2 + q^3 & p^3 + p^2q + 2pq^2 \\ p^2q + 3pq^2 & p^3 + 2p^2q + q^3 \\ p^3 + 2pq^2 + q^3 & 3p^2q + pq^2 \\ 2p^2q + pq^2 + q^3 & p^3 + p^2q + 2pq^2 \end{bmatrix} \qquad ITM_{circ} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}$$

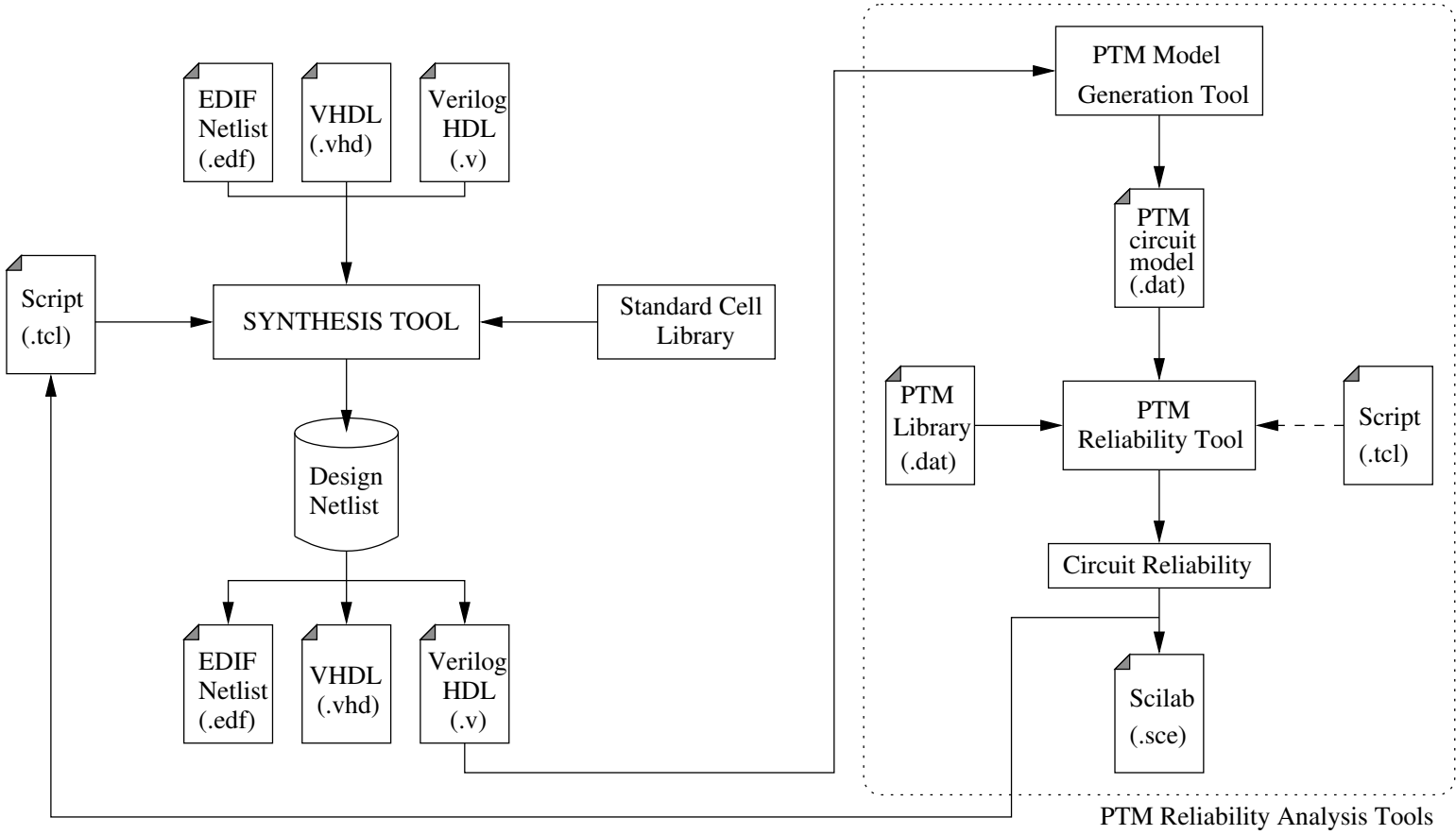
$$\begin{aligned} R = {} & a_0b_0c_0\left(2p^2q + pq^2 + q^3\right) + a_0b_0c_1\left(p^3 + 2p^2q + q^3\right) + \\ & a_0b_1c_0\left(2p^2q + pq^2 + q^3\right) + a_0b_1c_1\left(p^3 + 2p^2q + q^3\right) + \\ & a_1b_0c_0\left(2p^2q + pq^2 + q^3\right) + a_1b_0c_1\left(p^3 + 2p^2q + q^3\right) + \\ & a_1b_1c_0\left(p^3 + 2pq^2 + q^3\right) + a_1b_1c_1\left(2p^2q + pq^2 + q^3\right) \end{aligned}$$

System versus Component Reliability



$$P(a_i b_i c_i) = \frac{1}{8} \Rightarrow R_{circ} = q^3 + \frac{3q^2}{4}(-q+1) + \frac{7q}{4}(-q+1)^2 + \frac{1}{2}(-q+1)^3$$

PTM Computing Flow



PTM – Conclusions

- Accurate reliability assessment
- Single and multiple faults
- PTM's size increases exponentially with the number of inputs and outputs
- Intractable computing times and memory storage needs for medium size circuits.
- Scalability problem

Probabilistic Binomial Reliability (PBR)

Fault State

Definition

Let be the set of n components in a system

$$\mathbb{C} = \{c_0, c_1, c_2, \cdots, c_{n-1}\}$$

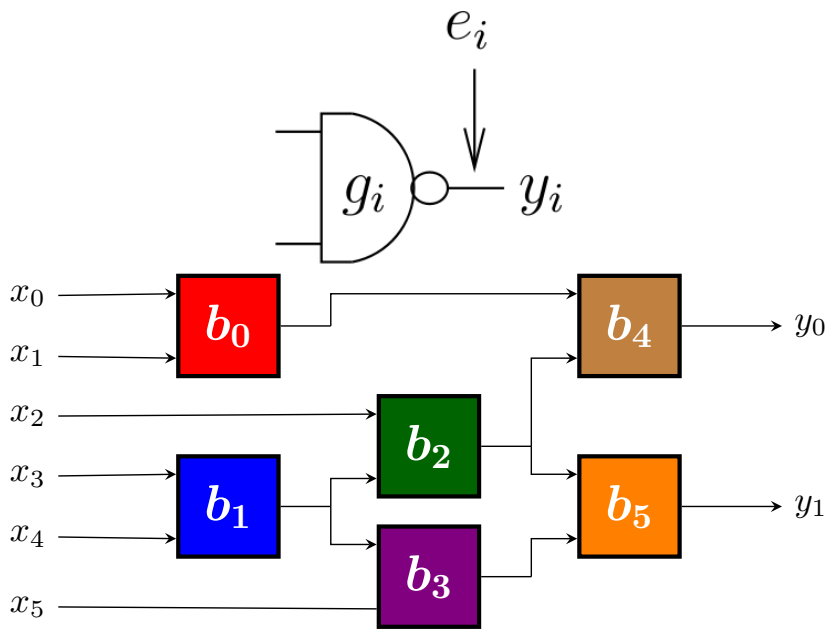
The **fault state of a component** i is defined as

$$f_i = \begin{cases} 0 & \text{if the component } i \text{ is fault-free} \\ 1 & \text{if the component } i \text{ is faulty} \end{cases}$$

for $i = 0, 1, 2, \cdots, n-1$.

Fault Modelling

- $f_i = 1$ inverts expected c_i output
- Remark: a component c_i can be a simple gate g_i or a block of gates $b_i = 1$



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System Fault State

Definition

The **system fault state** is defined as

$$\mathbf{f} = f_{n-1} \cdots f_2 f_1 f_0$$

The set of possible $\mathbf{f}$ is

$$\mathbb{F} = \{\mathbf{f}_0, \mathbf{f}_1, \cdots, \mathbf{f}_r, \cdots, \mathbf{f}_{2^n-1}\}$$

where $\mathbf{f}_0$ corresponds to all components fault-free.

- $\mathbb{F}_{n:k}$ is a subset of $\mathbb{F}$ of all $\mathbf{f}_r$ with k 1's
- $\mathbb{F}_{n:0} = \{\mathbf{f}_0\}$

System Input and Output

Definition

The system input is defined as

$$\mathbf{x} = x_{m-1} \cdots x_2 x_1 x_0 \text{ where } x_i \in \{0, 1\}$$

The set of possible $\mathbf{x}$ is

$$\mathbb{X} = \{\mathbf{x}_0, \mathbf{x}_1, \cdots, \mathbf{x}_v, \cdots, \mathbf{x}_{2^m-1}\}$$

where $\mathbf{x}_0$ means $x_i = 0$ for all $i = 0, 1, 2, \cdots, m-1$.
The system output is defined as

$$\mathbf{y} = y_{s-1} \cdots y_2 x_1 y_0 \text{ where } y_i \in \{0, 1\}$$

Check Function

Definition

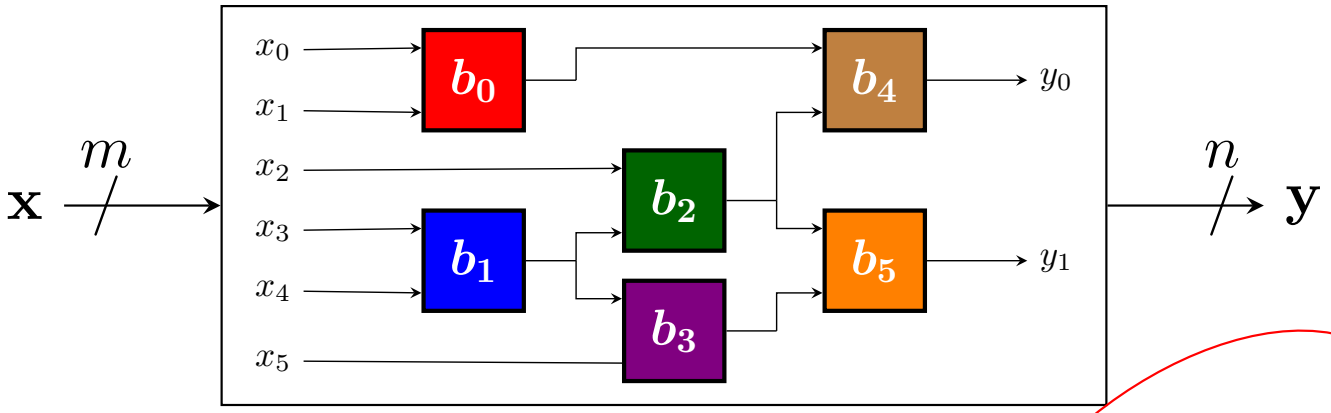
The check system function is defined as

$$h(v,r) = \begin{cases} 1, & \mathbf{y}(\mathbf{x}_v, \mathbf{f}_0) = \mathbf{y}(\mathbf{x}_v, \mathbf{f}_r) \\ 0, & \text{otherwise} \end{cases}$$

or

$$h(v,r) = \overline{\mathbf{y}(\mathbf{x}_v, \mathbf{f}_0) \oplus \mathbf{y}(\mathbf{x}_v, \mathbf{f}_r)}$$

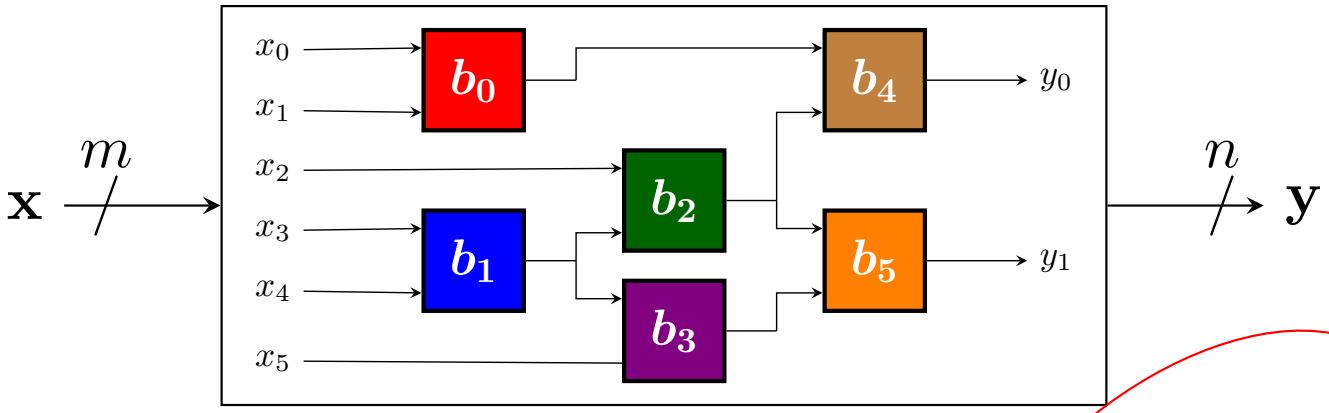
Reliability Calculation with PBR



$$R = \sum_{r=0}^{2^n-1} P\{\mathbf{f}_r\} \sum_{v=0}^{2^m-1} P\{\mathbf{x}_v\} \overbrace{(\mathbf{y}(\mathbf{x}_v, \mathbf{f}_0) = \mathbf{y}(\mathbf{x}_v, \mathbf{f}_r)}^{h(v,r)}$$

- Inputs relevance
- Technology & fault relevance
- Logical masking

Reliability Calculation with PBR



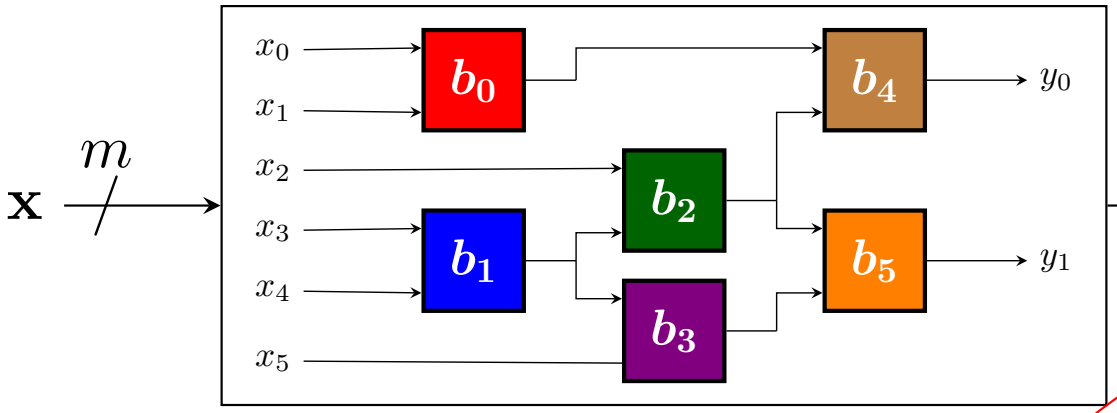
$$R = \sum_{r=0}^{2^n-1} P\{\mathbf{f}_r\} \sum_{v=0}^{2^m-1} P\{\mathbf{x}_v\} \overbrace{(\mathbf{y}(\mathbf{x}_v, \mathbf{f}_0) = \mathbf{y}(\mathbf{x}_v, \mathbf{f}_r)}$$

$h(v, r)$

- Inputs relevance
- Technology & fault relevance
- Logical masking

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Reliability Calculation with PBR

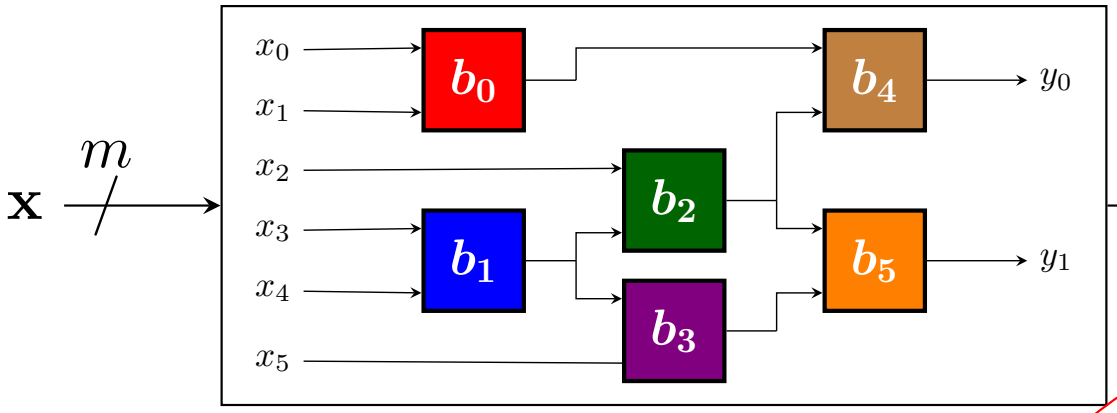


$$R = \sum_{r=0}^{2^n-1} P\{\mathbf{f}_r\} \sum_{v=0}^{2^m-1} P\{\mathbf{x}_v\} \overbrace{(\mathbf{y}(\mathbf{x}_v, \mathbf{f}_0) = \mathbf{y}(\mathbf{x}_v, \mathbf{f}_r)}^{h(v,r)}$$

- Inputs relevance
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Reliability Calculation with PBR



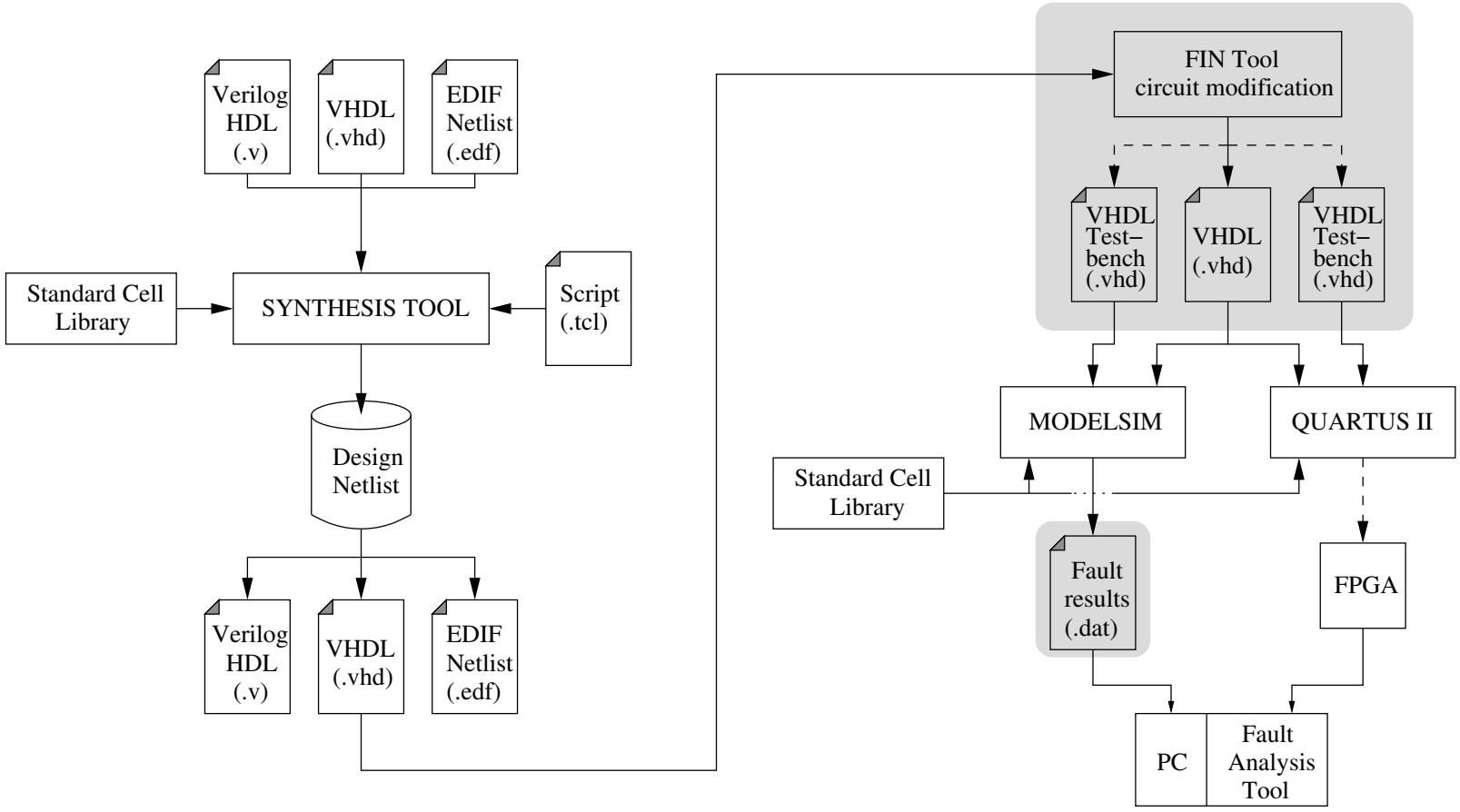
$$R = \sum_{r=0}^{2^n-1} P\{\mathbf{f}_r\} \sum_{v=0}^{2^m-1} P\{\mathbf{x}_v\} \overbrace{(\mathbf{y}(\mathbf{x}_v, \mathbf{f}_0) = \mathbf{y}(\mathbf{x}_v, \mathbf{f}_r)}$$

$h(v, r)$

- Inputs relevance
- Technology & fault relevance
- Logical masking

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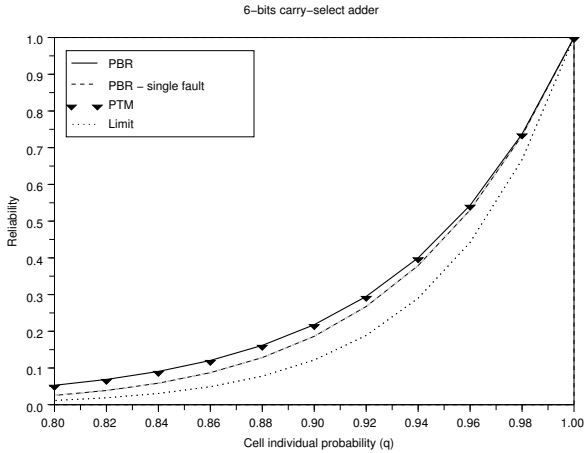
PBR Computing Flow



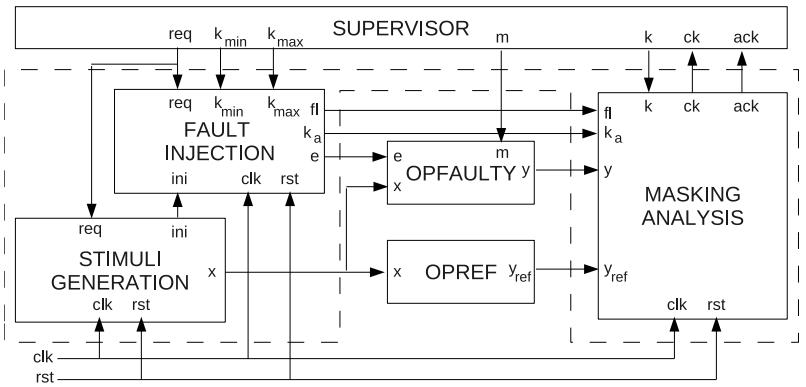
Simulation or FPGA emulation!

Accuracy and Scalability Issues

Relaxed implementaton and HW acceleration



Fault injection/analisys implemented by FPGA



PBR – Conclusions

- Accurate exhaustive approach
  - Gives exact results as PTM model. Requires less memory than PTM.
  - Takes into account all possible inputs and error configurations. Takes into account reconvergent fanouts
- Efficient tradeoffs related to accuracy and exhaustivity can be found
  - Relaxed version of PBR model giving exact (useful) results
- Suitable to be integrated in the design flow
  - Reliability metric becomes a control parameter for the synthesis.
- HW acceleration with FIFA platform is cost and time efficient

Signal Probability Analysis (SPR)

- Fault prone signals s are in one of four different states
- The possible states and respective probabilities are represented in two 2 by 2 matrices

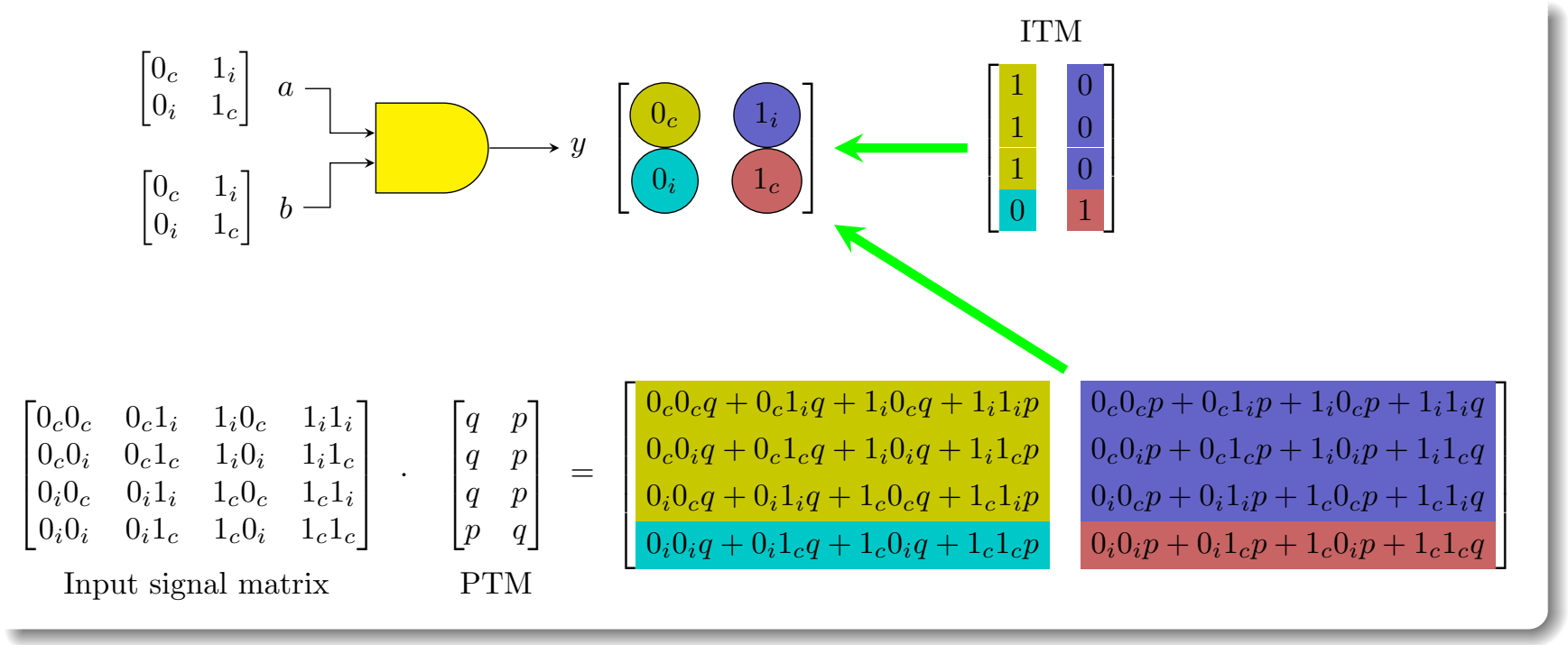
$$s = \begin{bmatrix} 0_c & 1_i \\ 0_i & 1_c \end{bmatrix} \text{ and } P\{s\} = \begin{bmatrix} P\{0_c\} & P\{1_i\} \\ P\{0_i\} & P\{1_c\} \end{bmatrix} \quad (2)$$

$$P\{0_c\} + P\{0_i\} + P\{1_i\} + P\{1_c\} = 1 \quad (3)$$

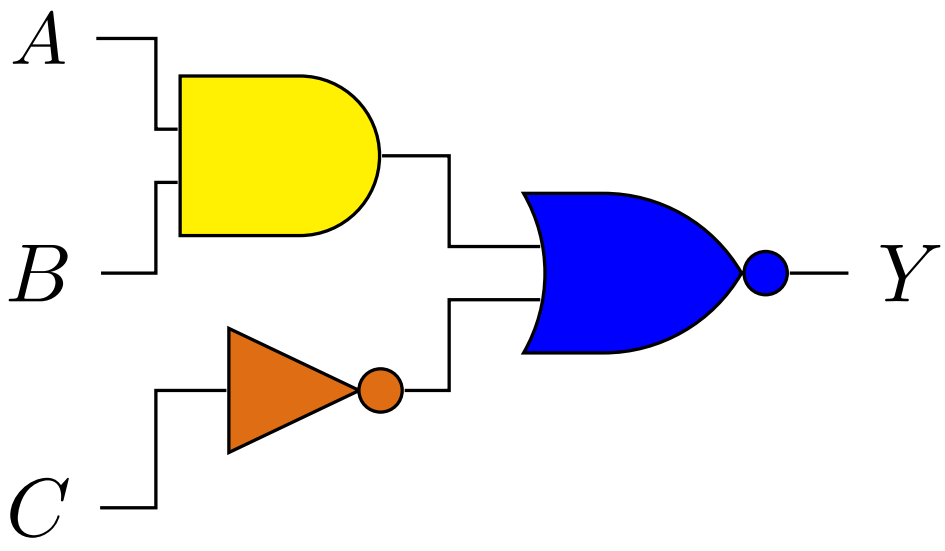
- The probability matrix $P(s)$ embeds the reliability information

$$R_s = P\{0_c\} + P\{1_c\} \quad (4)$$

Propagation of Signal Probabilities

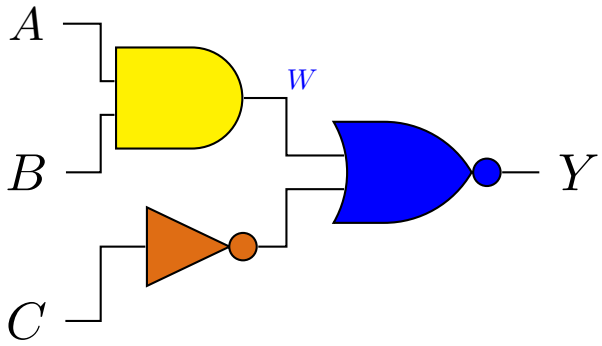


Example



$$p_{ij} = p \text{ and } q_{ij} = q \text{ for all } (i, j)$$

L1: Output of Gate NAND



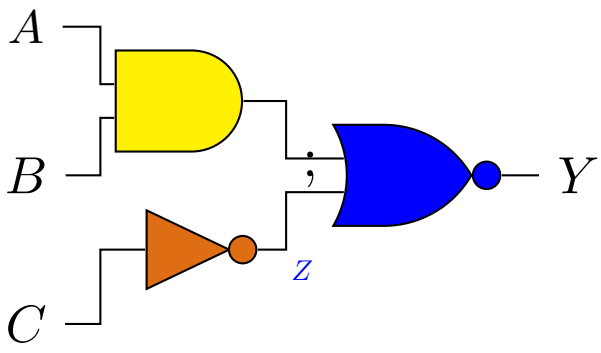
$$AB = \begin{pmatrix} a_{0c}b_{0c} & a_{0c}b_{1i} & a_{1i}b_{0c} & a_{1i}b_{1i} \\ a_{0c}b_{0i} & a_{0c}b_{1c} & a_{1i}b_{0i} & a_{1i}b_{1c} \\ a_{0i}b_{0c} & a_{0i}b_{1i} & a_{1c}b_{0c} & a_{1c}b_{1i} \\ a_{0i}b_{0i} & a_{0i}b_{1c} & a_{1c}b_{0i} & a_{1c}b_{1c} \end{pmatrix}$$

$$PTM = \begin{pmatrix} a_{0c}b_{0c}q + a_{0c}b_{1i}q + a_{1i}b_{0c}q + a_{1i}b_{1i}p & a_{0c}b_{0c}p + a_{0c}b_{1i}p + a_{1i}b_{0c}p + a_{1i}b_{1i}q \\ a_{0c}b_{0i}q + a_{0c}b_{1c}q + a_{1i}b_{0i}q + a_{1i}b_{1c}p & a_{0c}b_{0i}p + a_{0c}b_{1c}p + a_{1i}b_{0i}p + a_{1i}b_{1c}q \\ a_{0i}b_{0c}q + a_{0i}b_{1i}q + a_{1c}b_{0c}q + a_{1c}b_{1i}p & a_{0i}b_{0c}p + a_{0i}b_{1i}p + a_{1c}b_{0c}p + a_{1c}b_{1i}q \\ a_{0i}b_{0i}q + a_{0i}b_{1c}q + a_{1c}b_{0i}q + a_{1c}b_{1c}p & a_{0i}b_{0i}p + a_{0i}b_{1c}p + a_{1c}b_{0i}p + a_{1c}b_{1c}q \end{pmatrix}$$

$$\begin{aligned} w_{0c} &= a_{0c}b_{0c}q + a_{0c}b_{0i}q + a_{0c}b_{1c}q + a_{0c}b_{1i}q + a_{0i}b_{0c}q + a_{0i}b_{1i}q + \\ &\quad a_{1c}b_{0c}q + a_{1c}b_{1i}p + a_{1i}b_{0c}q + a_{1i}b_{0i}q + a_{1i}b_{1c}p + a_{1i}b_{1i}p \\ w_{0i} &= a_{0i}b_{0i}q + a_{0i}b_{1c}q + a_{1c}b_{0i}q + a_{1c}b_{1c}p \\ w_{1c} &= a_{0i}b_{0i}p + a_{0i}b_{1c}p + a_{1c}b_{0i}p + a_{1c}b_{1c}q \\ w_{1i} &= a_{0c}b_{0c}p + a_{0c}b_{0i}p + a_{0c}b_{1c}p + a_{0c}b_{1i}p + a_{0i}b_{0c}p + a_{0i}b_{1i}p + \\ &\quad a_{1c}b_{0c}p + a_{1c}b_{1i}q + a_{1i}b_{0c}p + a_{1i}b_{0i}p + a_{1i}b_{1c}q + a_{1i}b_{1i}q \end{aligned}$$

$$W = \begin{pmatrix} w_{0c} & w_{1i} \\ w_{0i} & w_{1c} \end{pmatrix}$$

L1: Output of Gate NOT



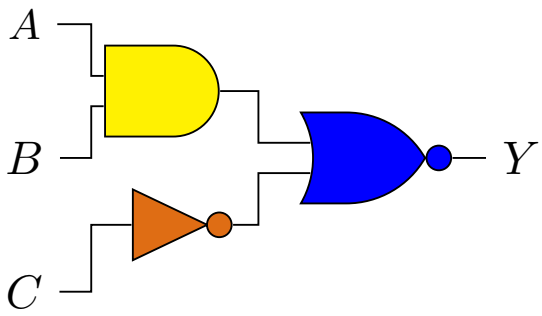
$$C = \begin{pmatrix} c_{0c} & c_{1i} \\ c_{0i} & c_{1c} \end{pmatrix}$$

$$PTM = \begin{pmatrix} c_{0c}p + c_{1i}q & c_{0c}q + c_{1i}p \\ c_{0i}p + c_{1c}q & c_{0i}q + c_{1c}p \end{pmatrix}$$

$$\begin{aligned} z_{0c} &= c_{0i}p + c_{1c}q \\ z_{0i} &= c_{0c}p + c_{1i}q \\ z_{1c} &= c_{0c}q + c_{1i}p \\ z_{1i} &= c_{0i}q + c_{1c}p \end{aligned}$$

$$Z = \begin{pmatrix} c_{0i}p + c_{1c}q & c_{0i}q + c_{1c}p \\ c_{0c}p + c_{1i}q & c_{0c}q + c_{1i}p \end{pmatrix}$$

L2: Output of Gate NOR



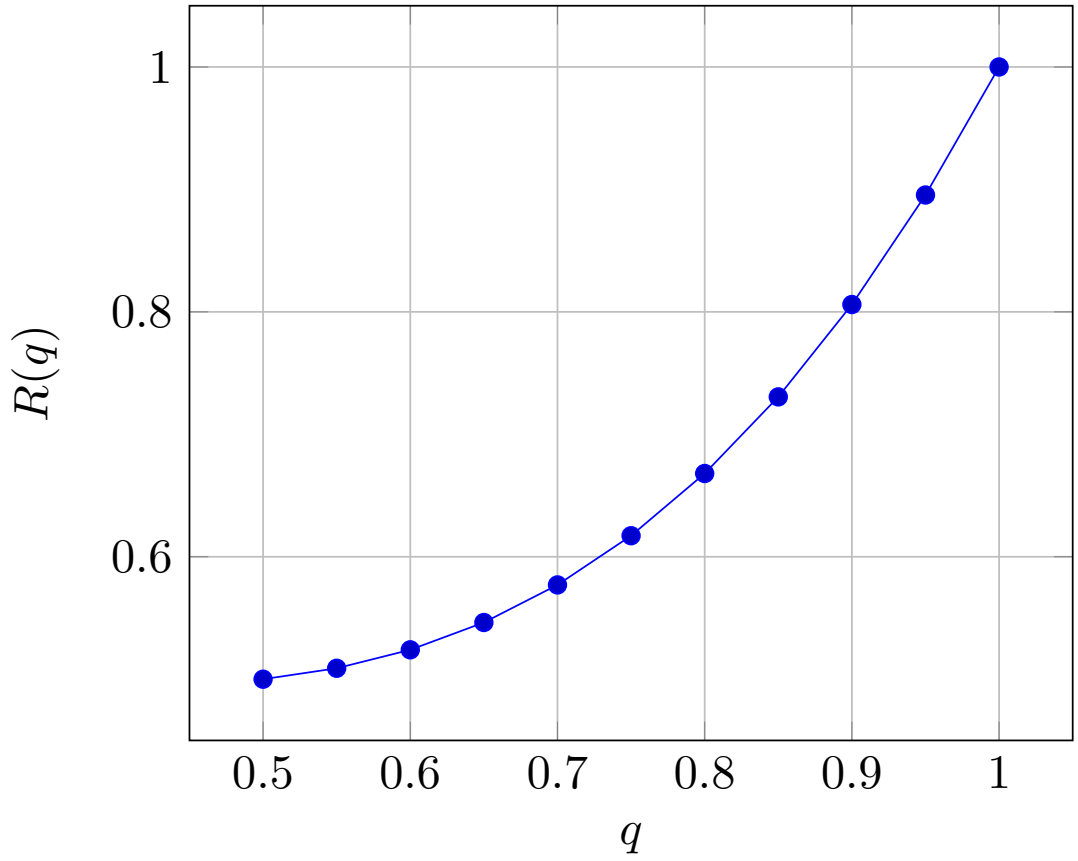
$$WZ = \begin{pmatrix} 3q^2/8 & 3q\left(-q+1\right)/8 & 3q\left(-q+1\right)/8 & 3\left(-q+1\right)^2/8 \\ 3q\left(-q+1\right)/8 & 3q^2/8 & 3\left(-q+1\right)^2/8 & 3q\left(-q+1\right)/8 \\ q\left(-q+1\right)/8 & \left(-q+1\right)^2/8 & q^2/8 & q\left(-q+1\right)/8 \\ \left(-q+1\right)^2/8 & q\left(-q+1\right)/8 & q\left(-q+1\right)/8 & q^2/8 \end{pmatrix}$$

$$PTM = \begin{pmatrix} \frac{3p}{8}q^2 + \frac{3q^2}{4}\left(-q+1\right) + \frac{3q}{8}\left(-q+1\right)^2 & \frac{3p}{4}q\left(-q+1\right) + \frac{3p}{8}\left(-q+1\right)^2 + \frac{3q^3}{8} \\ \frac{3p}{8}q\left(-q+1\right) + \frac{3q^3}{8} + \frac{3q^2}{8}\left(-q+1\right) + \frac{3q}{8}\left(-q+1\right)^2 & \frac{3p}{8}q^2 + \frac{3p}{8}q\left(-q+1\right) + \frac{3p}{8}\left(-q+1\right)^2 + \frac{3q^2}{8}\left(-q+1\right) \\ \frac{pq}{8}\left(-q+1\right) + \frac{q^3}{8} + \frac{q^2}{8}\left(-q+1\right) + \frac{q}{8}\left(-q+1\right)^2 & \frac{pq^2}{8} + \frac{pq}{8}\left(-q+1\right) + \frac{p}{8}\left(-q+1\right)^2 + \frac{q^2}{8}\left(-q+1\right) \\ \frac{p}{8}\left(-q+1\right)^2 + \frac{q^3}{8} + \frac{q^2}{4}\left(-q+1\right) & \frac{pq^2}{8} + \frac{pq}{4}\left(-q+1\right) + \frac{q}{8}\left(-q+1\right)^2 \end{pmatrix}$$

$$\begin{aligned} y_{0c} &= \frac{5q^3}{8} + \frac{3q^2}{4}\left(-q+1\right) + q\left(-q+1\right)^2 + \frac{1}{8}\left(-q+1\right)^3 \\ y_{0i} &= \frac{9q^2}{8}\left(-q+1\right) + \frac{3q}{8}\left(-q+1\right)^2 \\ y_{1c} &= \frac{3q^3}{8} + \frac{3q}{4}\left(-q+1\right)^2 + \frac{3}{8}\left(-q+1\right)^3 \\ y_{1i} &= \frac{9q^2}{8}\left(-q+1\right) + \frac{7q}{8}\left(-q+1\right)^2 + \frac{1}{2}\left(-q+1\right)^3 \end{aligned}$$

$$Y = \begin{pmatrix} y_{0c} & y_{1i} \\ y_{0i} & y_{1c} \end{pmatrix}$$

System versus Component Reliability



$$P(a_i b_i c_i) = \frac{1}{8} \Rightarrow R_{circ} = q^3 + \frac{3q^2}{4}(-q+1) + \frac{7q}{4}(-q+1)^2 + \frac{1}{2}(-q+1)^3$$

SPR Model

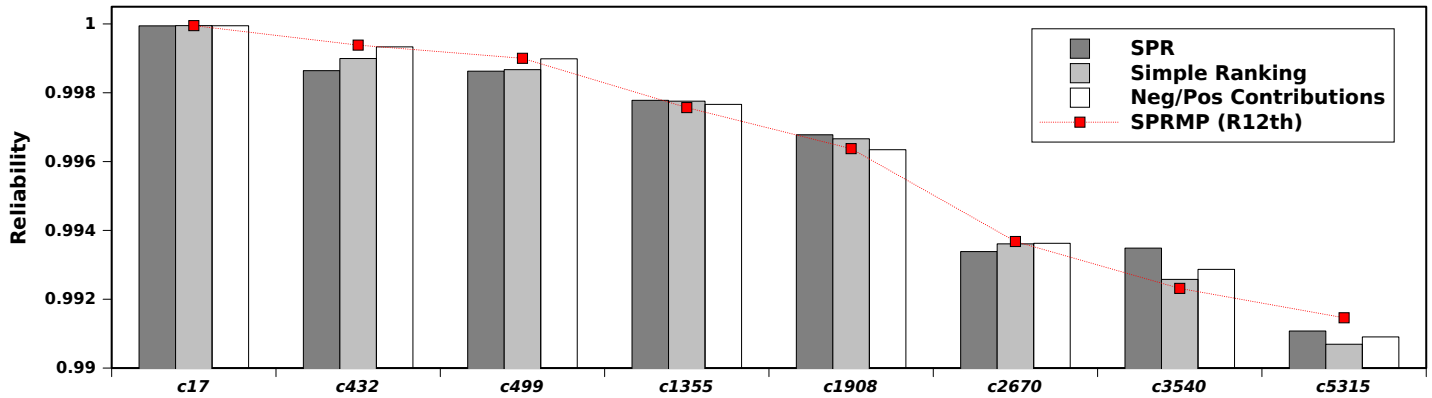
Dealing with the effect of reconvergence

- Dynamic Weighted Averaging Algorithm – DWAA (Linear complexity)
- Multi-pass Algorithm – MP (Linear to exponential complexity)
- Conditional Probability Method – CPM (Linear complexity)

First Order Fanout Analysis (SPR+)

Profile of the Fanout Nodes

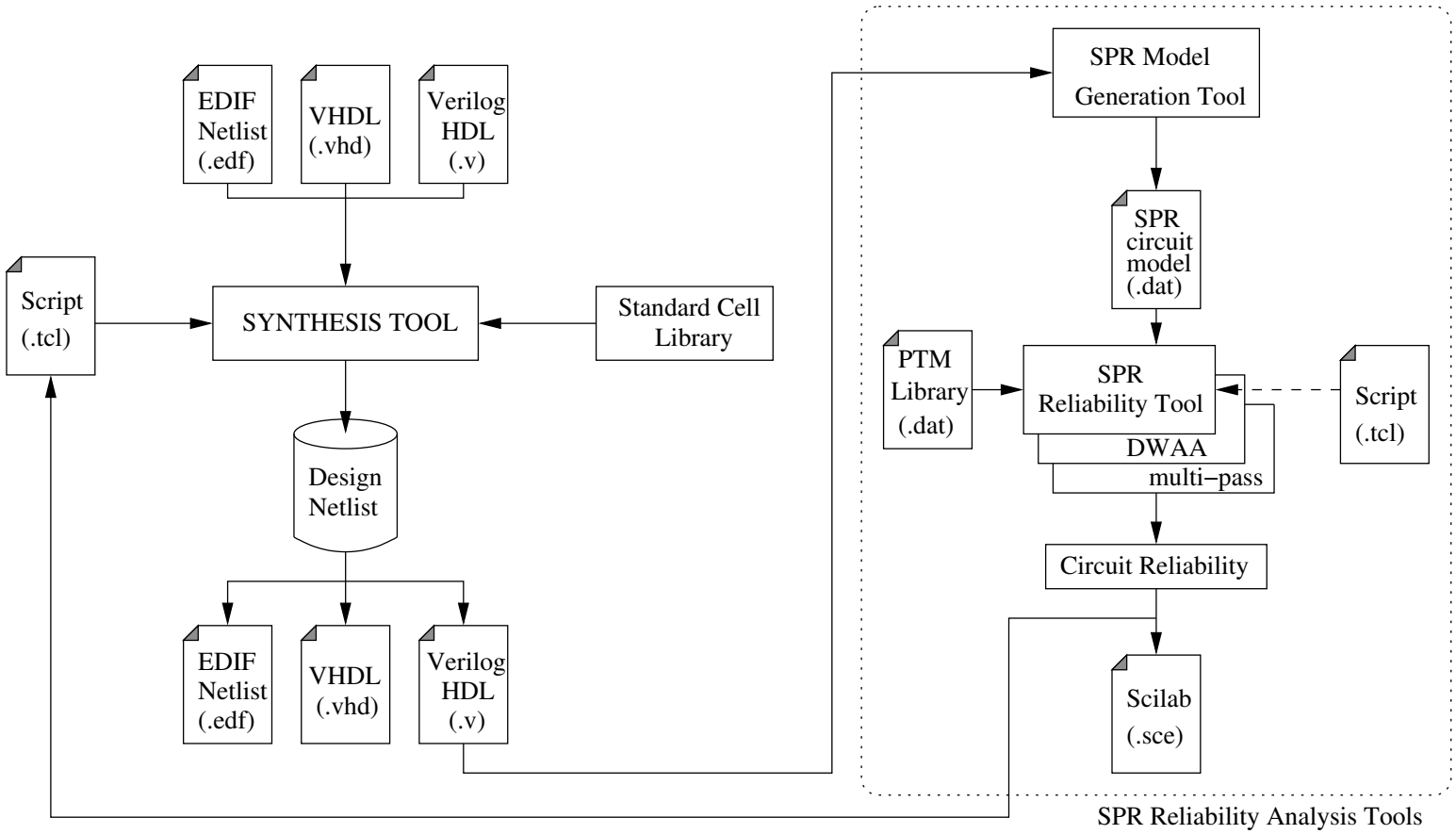
$$D(f) = |R_1(f) - R_0| \text{ and } f \text{ has } \begin{cases} \text{High impact,} & \text{if } D(f)/D_{max} > 0.8 \\ \text{Medium impact,} & \text{if } D(f)/D_{max} > 0.2 \\ \text{Low impact,} & \text{otherwise} \end{cases}$$



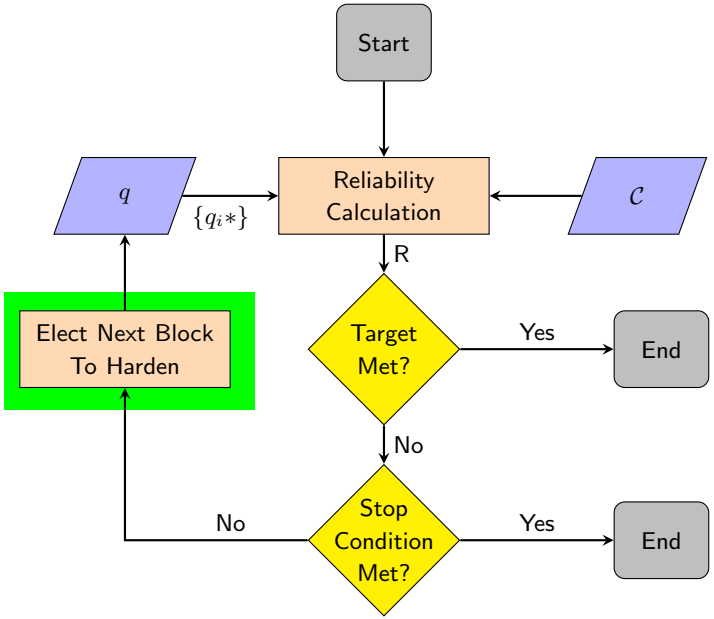
TELECOM Paris

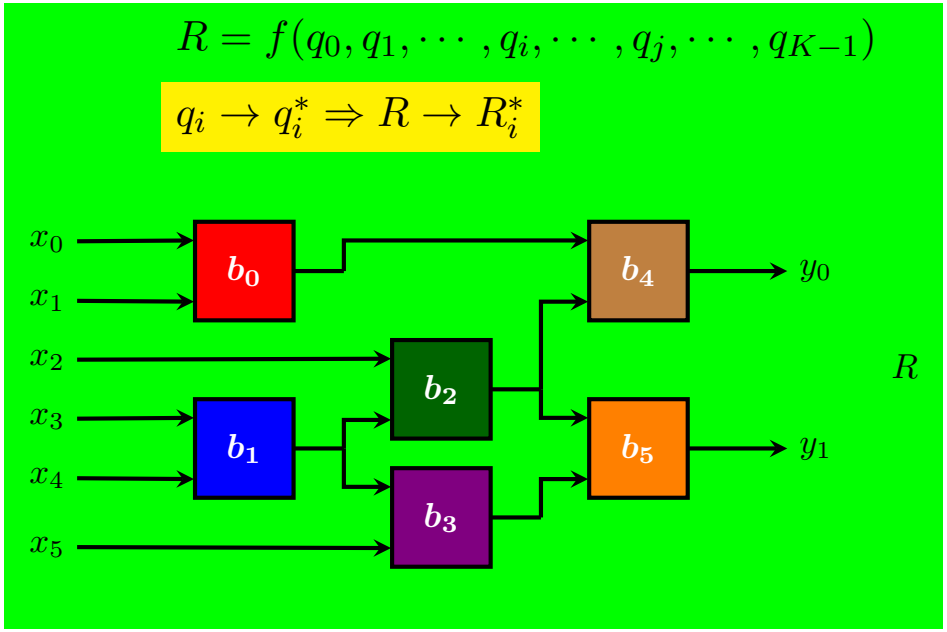
IP PARIS

SPR Computing Flow



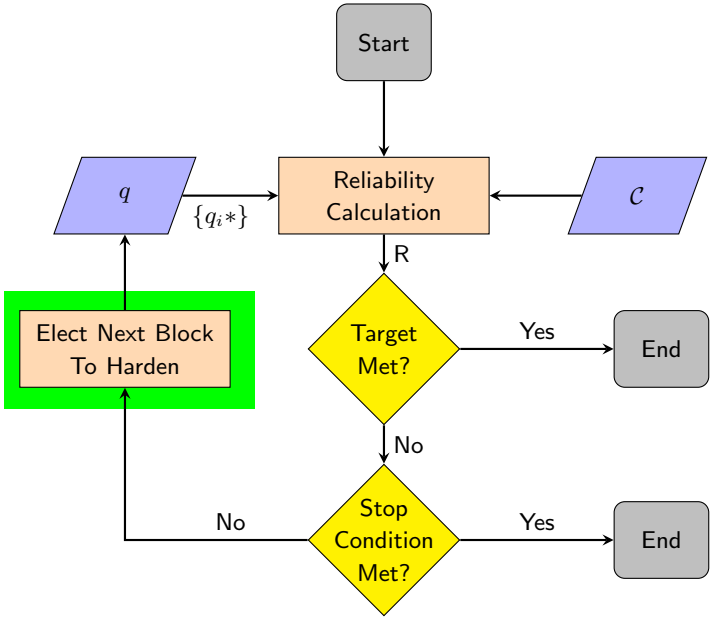
Selective Hardening

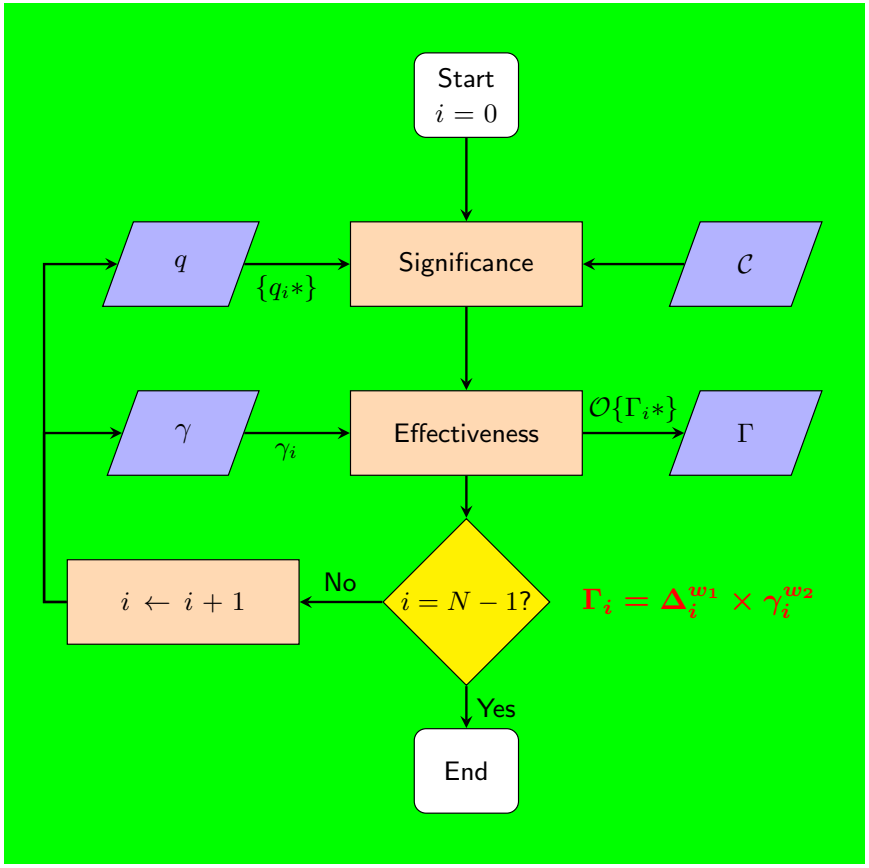




- Sensitivity to reliability change: $\sigma_i = \partial R / \partial q_i$
- Impact of reliability improvement $q_i \rightarrow q_i^*$: $\Delta_i = R_i^* - R$
- Easiness to harden: $\gamma = f(A, P, T)$

Selective Hardening





- New partial ordering is needed after hardening a block

Conclusions

- Reliability issues and challenges



- Need of cost-effective fault tolerant architectures
- Need of efficient assessment approaches

- Main issues of reliability assessment approaches
  - Scalability: accuracy, complexity
  - Design flow integration